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Course:

Operating System

Section:

BCS-5(B)

PROJECT REPORT

Contents

1. Introduction

This project demonstrates the implementation of essential network services including **FTP Server, Web Server, DNS Server, and DHCP Server** using **Ubuntu Linux** in a virtualized environment. The complete setup was implemented using **Oracle VirtualBox**.

The project starts from installing VirtualBox, setting up Ubuntu using an ISO file, configuring virtual machines, and finally deploying network services. Two network adapters (**NAT** and **Host-Only Adapter**) were used to simulate real-world networking.

2. Objectives

The objectives of this project are:

- To install and configure Oracle VirtualBox
- To install Ubuntu Linux using ISO file
- To create and configure virtual machines
- To implement FTP, Web, DNS, and DHCP servers
- To establish communication between server and client systems
- To gain practical experience in Linux networking

3. Tools and Technologies Used

- Oracle VirtualBox
- Ubuntu Linux ISO File
- Linux Terminal
- **Networking Services:** FTP, HTTP, DNS, DHCP
- **Network Adapters:** NAT & Host-Only

4. Methodology

The project was completed using the following methodology:

1. Installation of Oracle VirtualBox
2. Creation of two Ubuntu virtual machines
3. Configuration of network adapters (NAT and Host-Only)
4. Verification of connectivity using `ping`

5. Installation and configuration of servers (FTP, Web, DNS, DHCP)
6. Testing services from the client machine

5. VirtualBox Installation

Oracle VirtualBox was downloaded from the official website and installed on the host system. It provides a platform to run multiple operating systems simultaneously.

– Installation of Virtual Box and Ubunto:

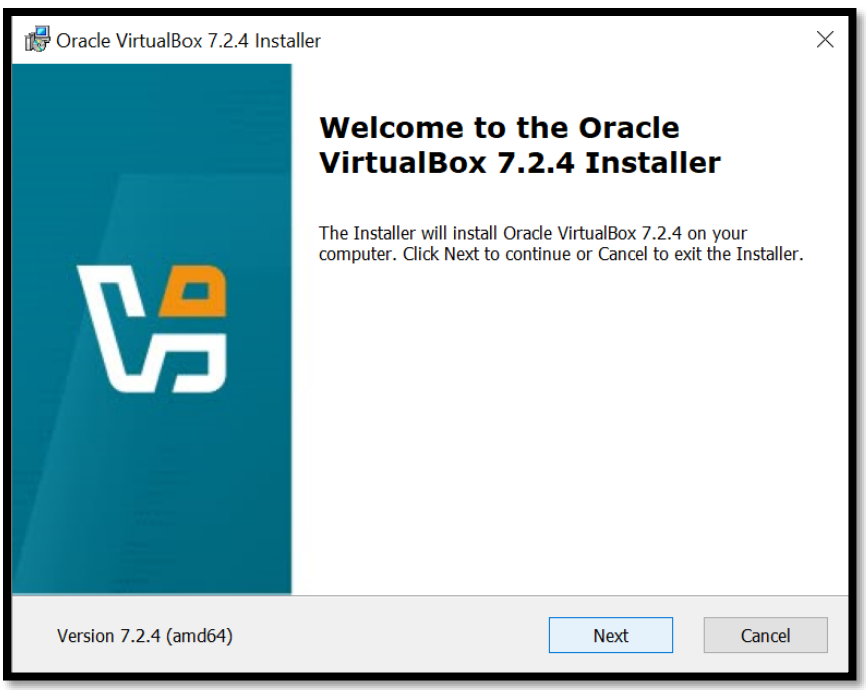
Step 1. The official Oracle VirtualBox website was visited to download the VirtualBox installer for the host operating system.

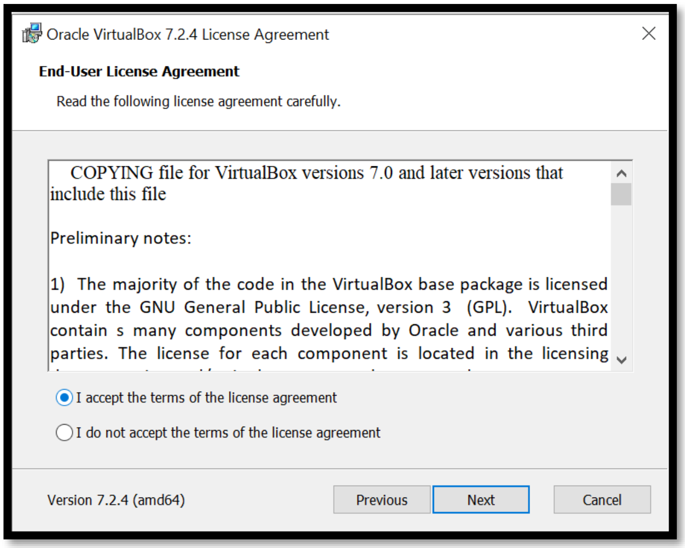
Step 2. After the download was completed, the installer file was opened to begin the installation process.

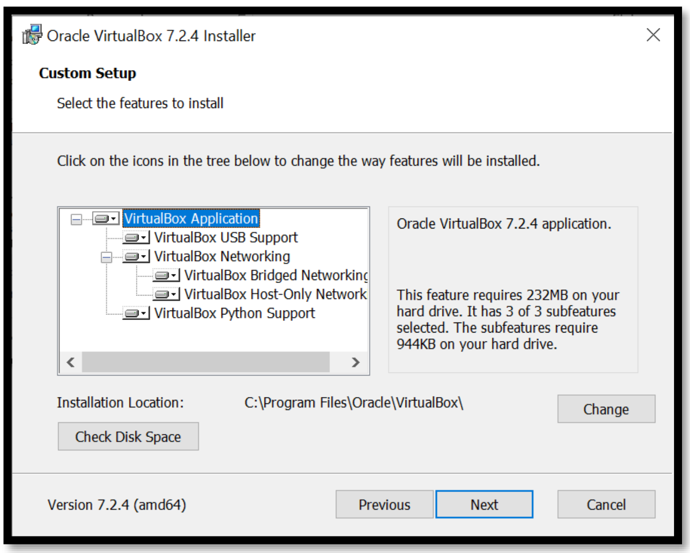
Step 3. The setup wizard was followed by clicking the **Next** button and keeping the default installation settings.

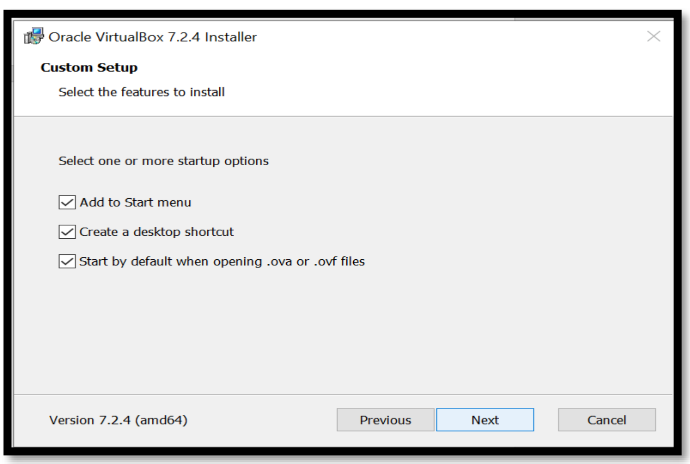
Step 4. Network and shortcut options were confirmed, and the installation was started.

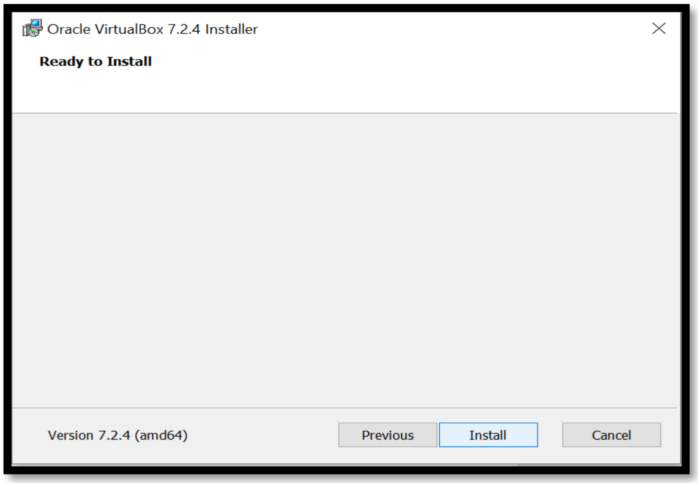
Step 5. Once the installation finished, Oracle VirtualBox was launched successfully and was ready for creating virtual machines.

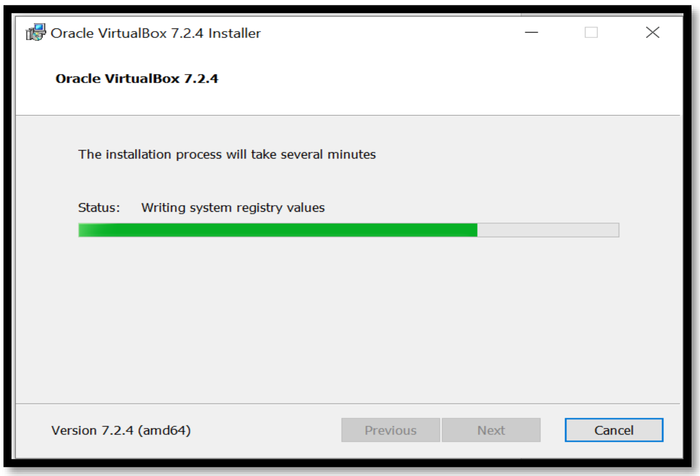


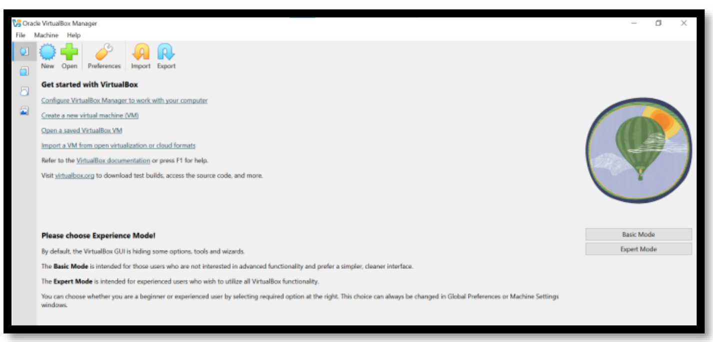

Oracle VirtualBox 7.2.4 License Agreement
End-User License Agreement
Read the following license agreement carefully.
COPYING file for VirtualBox versions 7.0 and later versions that include this file
Preliminary notes:
1) The majority of the code in the VirtualBox base package is licensed under the GNU General Public License, version 3 (GPL). VirtualBox contain s many components developed by Oracle and various third parties. The license for each component is located in the licensing
I accept the terms of the license agreement
I do not accept the terms of the license agreement
Version 7.2.4 (amd64)
Previous
Next
Cancel


Oracle VirtualBox 7.2.4 Installer
Custom Setup
Select the features to install
Click on the icons in the tree below to change the way features will be installed.
VirtualBox Application
VirtualBox USB Support
VirtualBox Networking
VirtualBox Bridged Networking
VirtualBox Host-Only Network
VirtualBox Python Support
Oracle VirtualBox 7.2.4 application.
This feature requires 232MB on your hard drive. It has 3 of 3 subfeatures selected. The subfeatures require 944KB on your hard drive.
Installation Location:
C:\Program Files\Oracle\VirtualBox\
Change
Check Disk Space
Version 7.2.4 (amd64)
Previous
Next
Cancel


Oracle VirtualBox 7.2.4 Installer
Custom Setup
Select the features to install
Select one or more startup options
Add to Start menu
Create a desktop shortcut
Start by default when opening .ova or .ovf files
Version 7.2.4 (amd64)
Previous
Next
Cancel



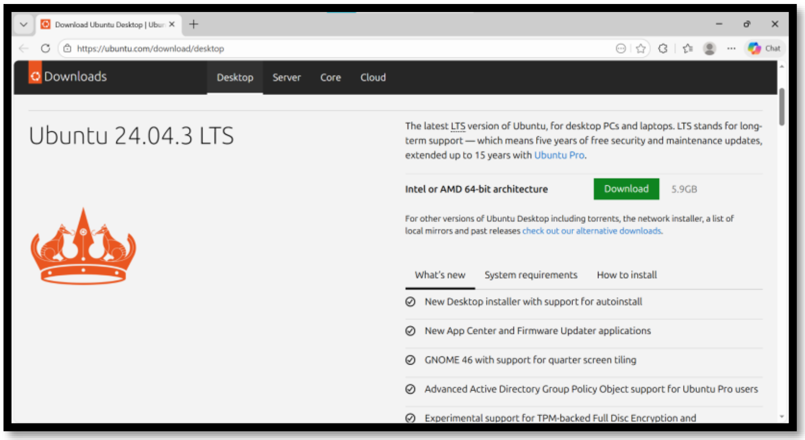




UBUNTU INSTALLATION:

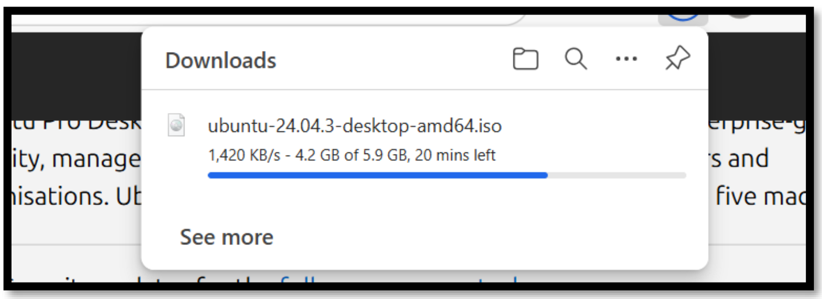
• **The Ubuntu Desktop ISO file was downloaded from the official Ubuntu site.**

Step 1. The official Ubuntu website was accessed to obtain the Ubuntu ISO file.



Step 2. The download process began and continued until completion.

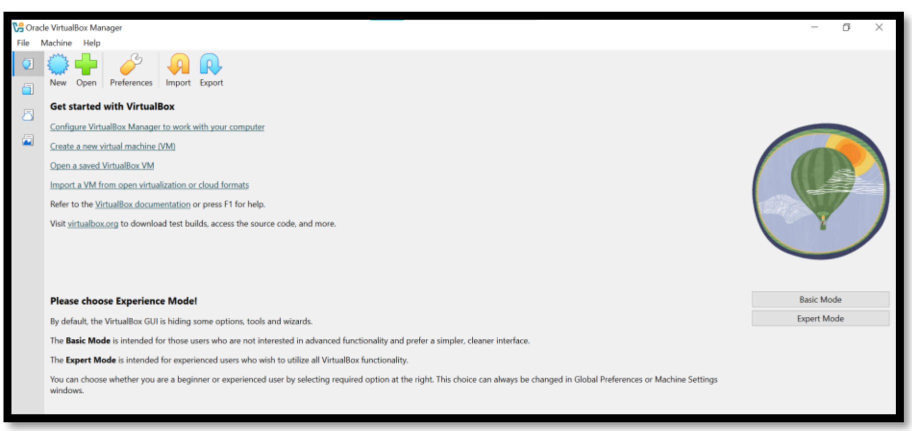
After the download was completed, the ISO file was saved in the Downloads directory.



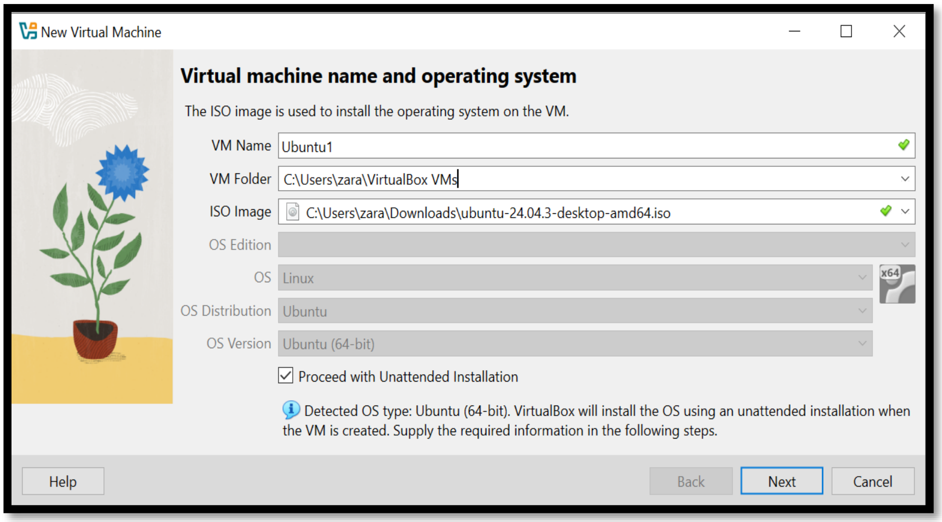
- A new virtual machine was created and Ubuntu was installed inside VirtualBox.

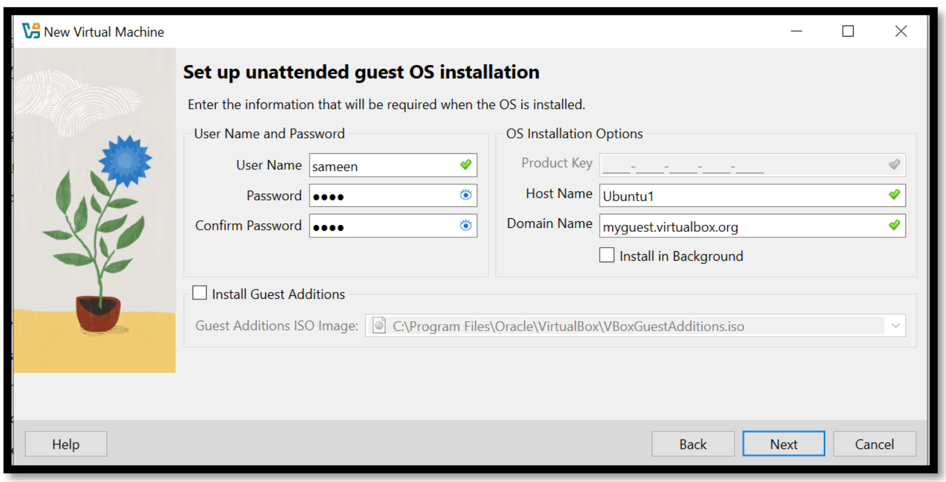
Creating Two Ubuntu Virtual Machines

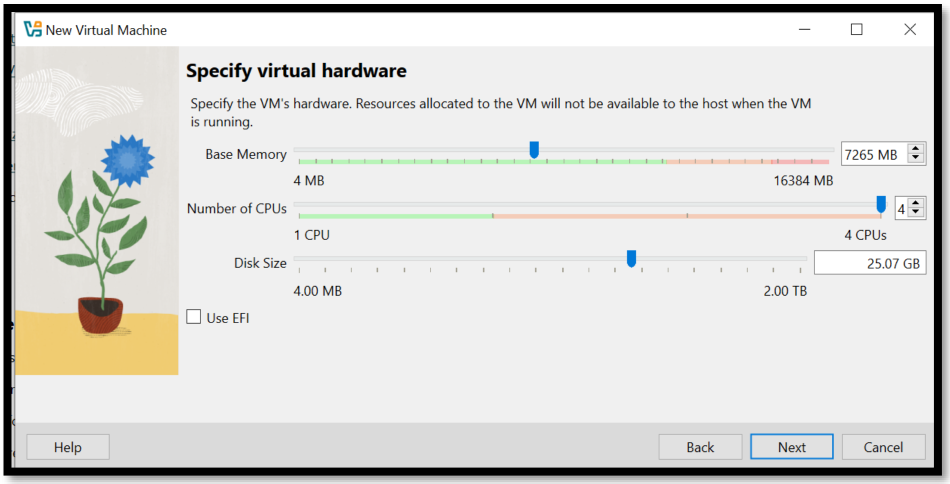
- **Step 1.** Oracle VirtualBox was launched and the option to create a new virtual machine was selected.

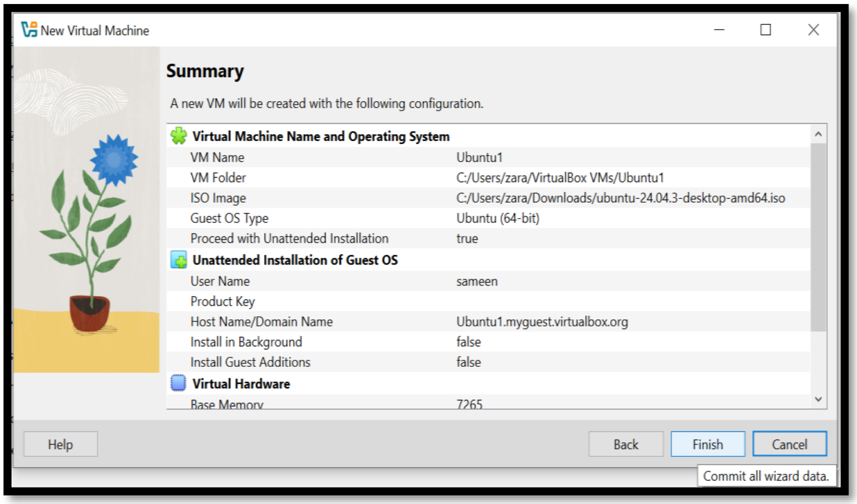


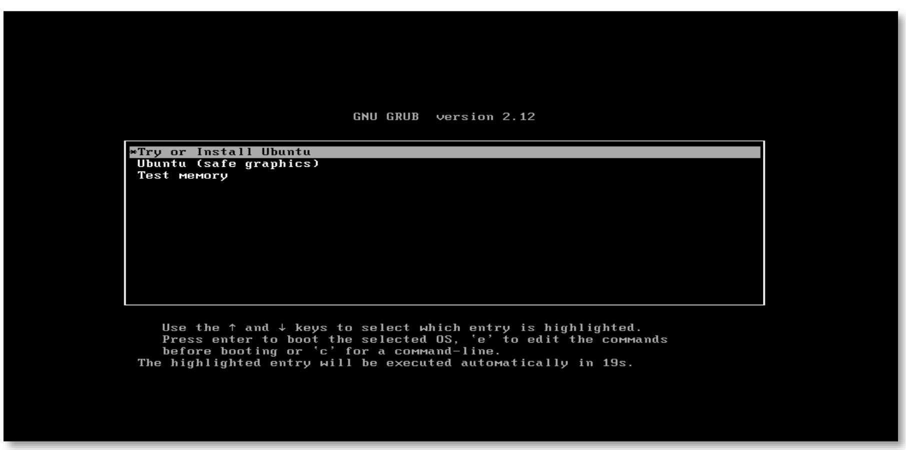
- **Step 2. The first virtual machine (Ubuntu 1) was created, and the Ubuntu operating system was installed.**
  The downloaded Ubuntu ISO file was selected as the installation source.

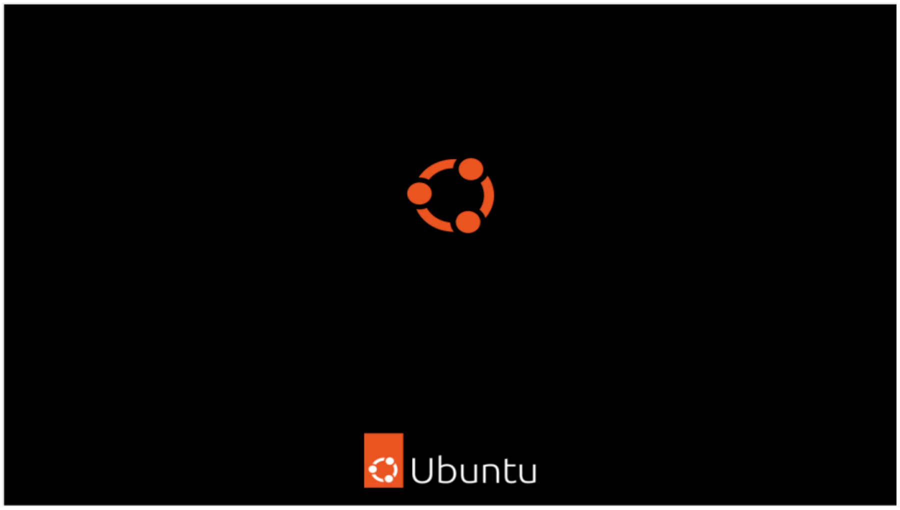


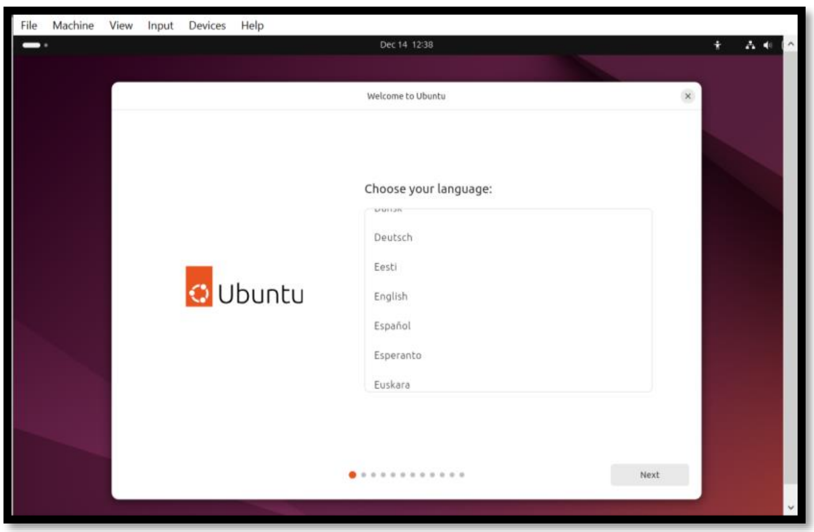


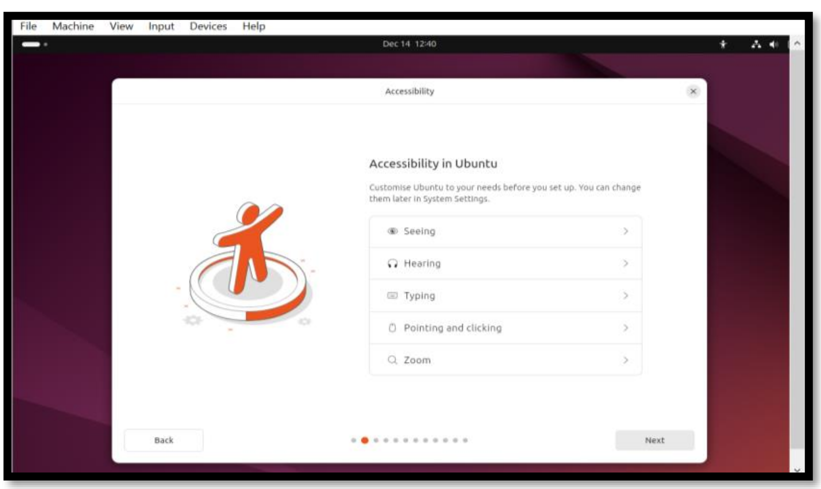

New Virtual Machine
Specify virtual hardware
Specify the VM's hardware. Resources allocated to the VM will not be available to the host when the VM is running.
Base Memory
7265 MB
4 MB
16384 MB
Number of CPUs
4
1 CPU
4 CPUs
Disk Size
25.07 GB
4.00 MB
2.00 TB
Use EFI
Help
Back
Next
Cancel

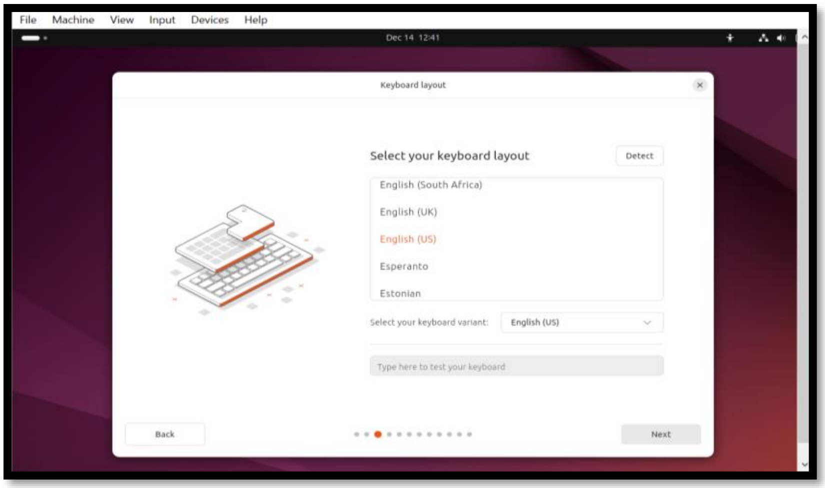

New Virtual Machine
Summary
A new VM will be created with the following configuration.
Virtual Machine Name and Operating System
VM Name Ubuntu1
VM Folder C:/Users/zara/VirtualBox VMs/Ubuntu1
ISO Image C:/Users/zara/Downloads/ubuntu-24.04.3-desktop-amd64.iso
Guest OS Type Ubuntu (64-bit)
Proceed with Unattended Installation true
Unattended Installation of Guest OS
User Name sameen
Product Key
Host Name/Domain Name Ubuntu1.myguest.virtualbox.org
Install in Background false
Install Guest Additions false
Virtual Hardware
Base Memory 7265
Help
Back
Finish
Cancel
Commit all wizard data.

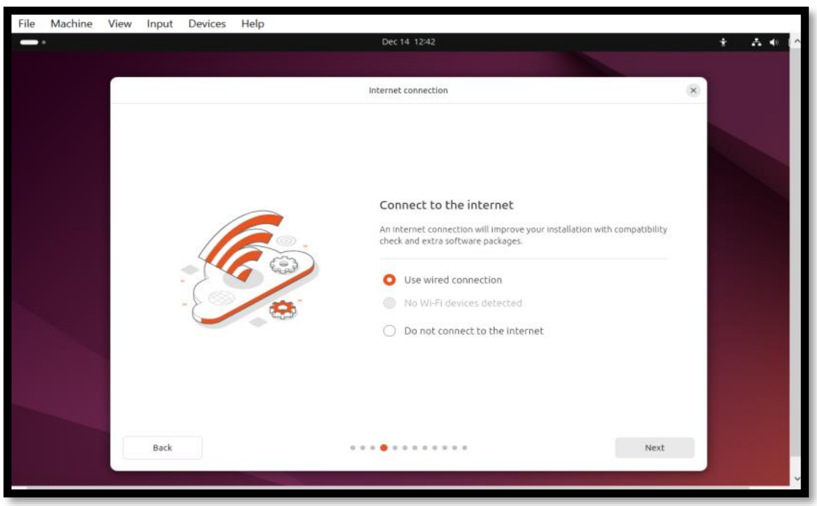

GNU GRUB version 2.12
*Try or Install Ubuntu
Ubuntu (safe graphics)
Test memory
Use the ↑ and ↓ keys to select which entry is highlighted.
Press enter to boot the selected OS, 'e' to edit the commands
before booting or 'c' for a command-line.
The highlighted entry will be executed automatically in 19s.

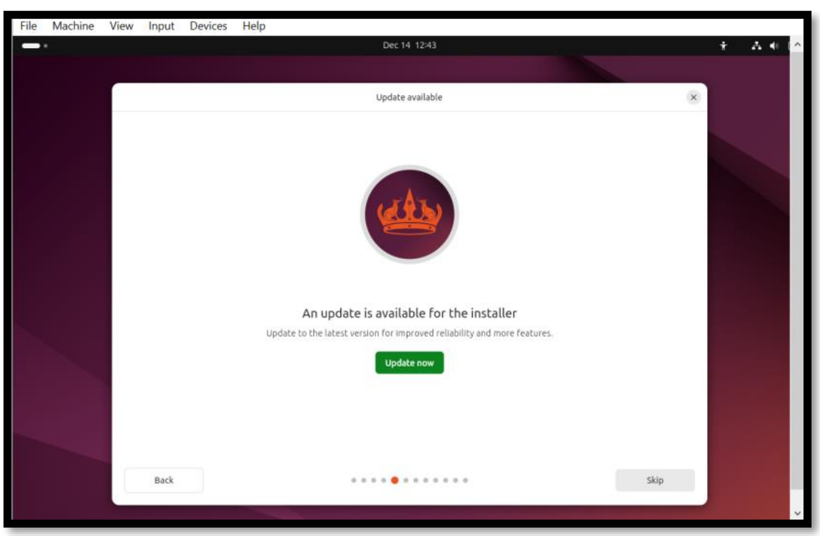

Ubuntu

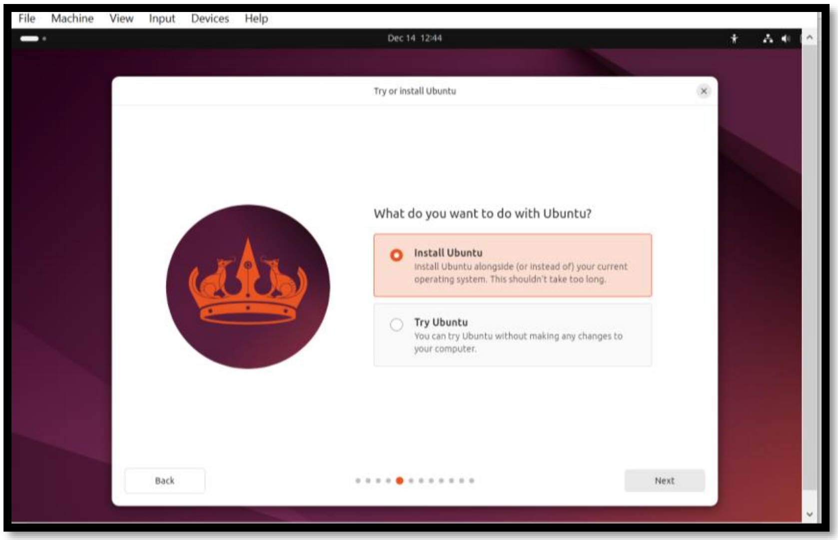

File
Machine
View
Input
Devices
Help
Dec 14 12:38
Welcome to Ubuntu
Ubuntu
Choose your language:
Deutsch
Eesti
English
Español
Esperanto
Euskara
Next

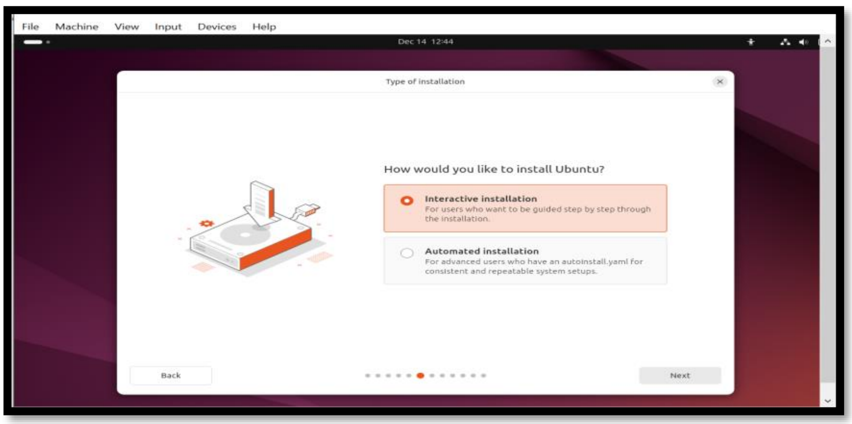

File
Machine
View
Input
Devices
Help
Dec 14 12:40
Accessibility
Accessibility in Ubuntu
Customise Ubuntu to your needs before you set up. You can change them later in System Settings.
Seeing
Hearing
Typing
Pointing and clicking
Zoom
Back
Next

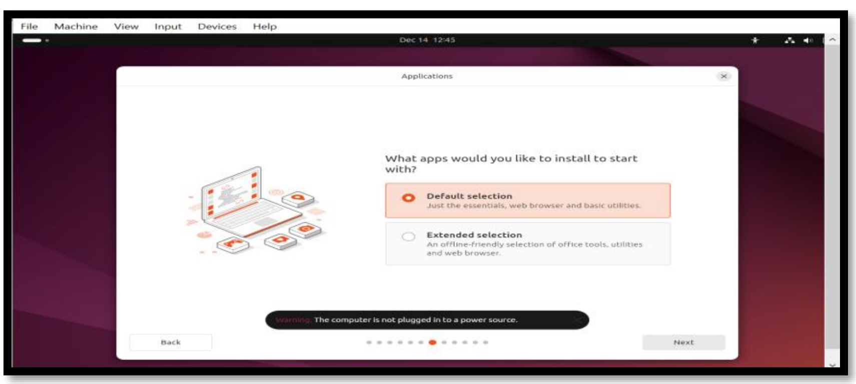

File Machine View Input Devices Help
Dec 14 12:41
Keyboard layout
Select your keyboard layout
Detect
English (South Africa)
English (UK)
English (US)
Esperanto
Estonian
Select your keyboard variant:
English (US)
Type here to test your keyboard
Back
Next

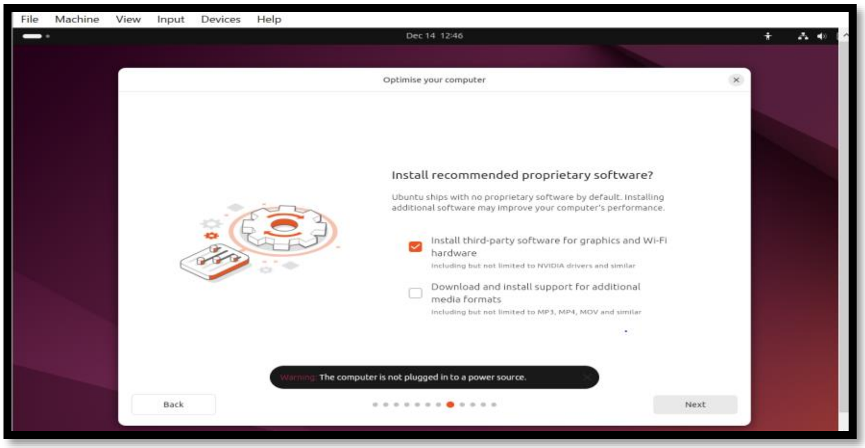

File Machine View Input Devices Help
Dec 14 12:42
Internet connection
Connect to the internet
An internet connection will improve your installation with compatibility check and extra software packages.
Use wired connection
No Wi-Fi devices detected
Do not connect to the internet
Back
Next

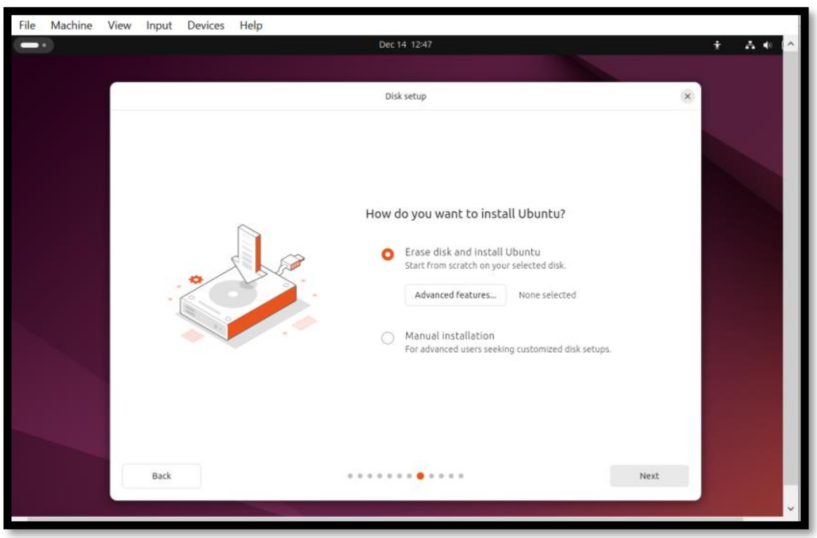

File Machine View Input Devices Help
Dec 14 12:43
Update available
An update is available for the installer
Update to the latest version for improved reliability and more features.
Update now
Back
Skip

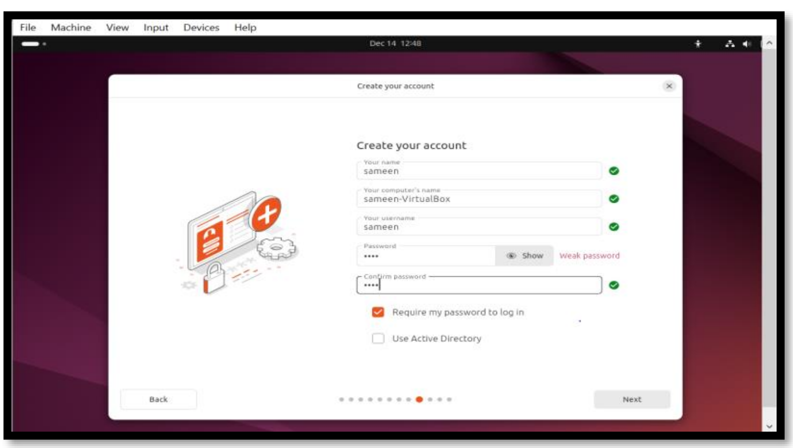

File Machine View Input Devices Help
Dec 14 12:44
Try or install Ubuntu
What do you want to do with Ubuntu?
Install Ubuntu
Install Ubuntu alongside (or instead of) your current operating system. This shouldn't take too long.
Try Ubuntu
You can try Ubuntu without making any changes to your computer.
Back
Next


File Machine View Input Devices Help
Dec 14 12:44
Type of installation
How would you like to install Ubuntu?
Interactive installation
For users who want to be guided step by step through the installation.
Automated installation
For advanced users who have an autoinstall.yaml for consistent and repeatable system setups.
Back
Next

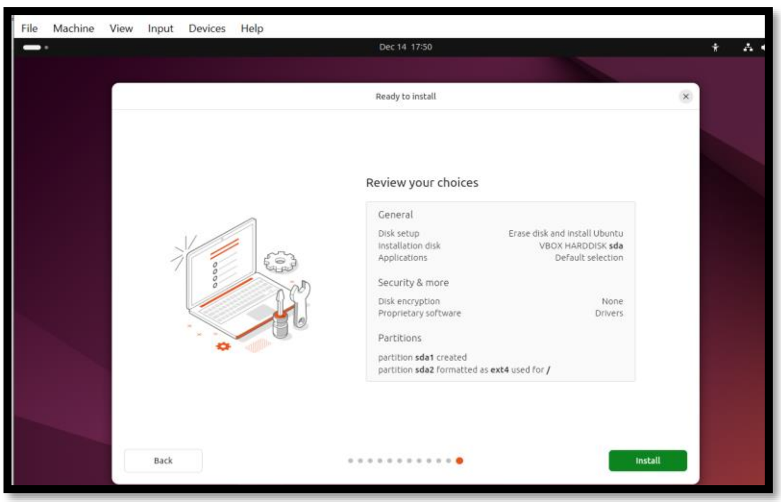

File Machine View Input Devices Help
Dec 14 12:45
Applications
What apps would you like to install to start with?
Default selection
Just the essentials, web browser and basic utilities.
Extended selection
An offline-friendly selection of office tools, utilities and web browser.
Warning: The computer is not plugged in to a power source.
Back
Next

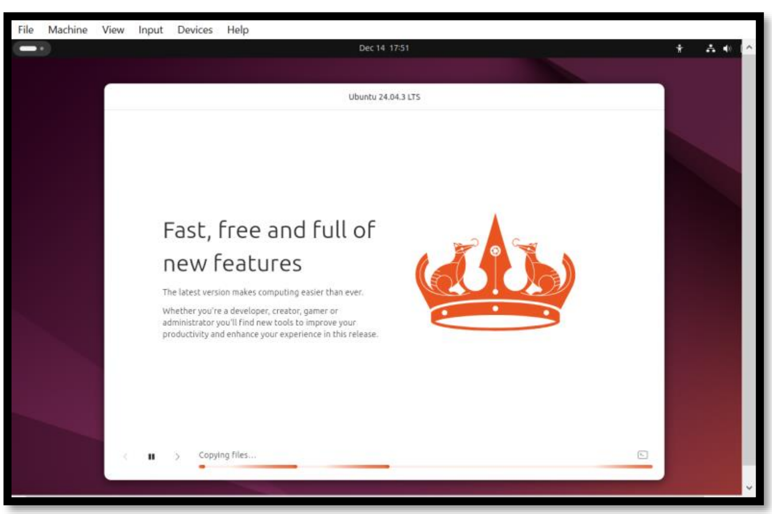

File Machine View Input Devices Help
Dec 14 12:46
Optimise your computer
Install recommended proprietary software?
Ubuntu ships with no proprietary software by default. Installing additional software may improve your computer's performance.
Install third-party software for graphics and Wi-Fi hardware
Including but not limited to NVIDIA drivers and similar
Download and install support for additional media formats
Including but not limited to MP3, MP4, MOV and similar
Warning: The computer is not plugged in to a power source.
Back
Next

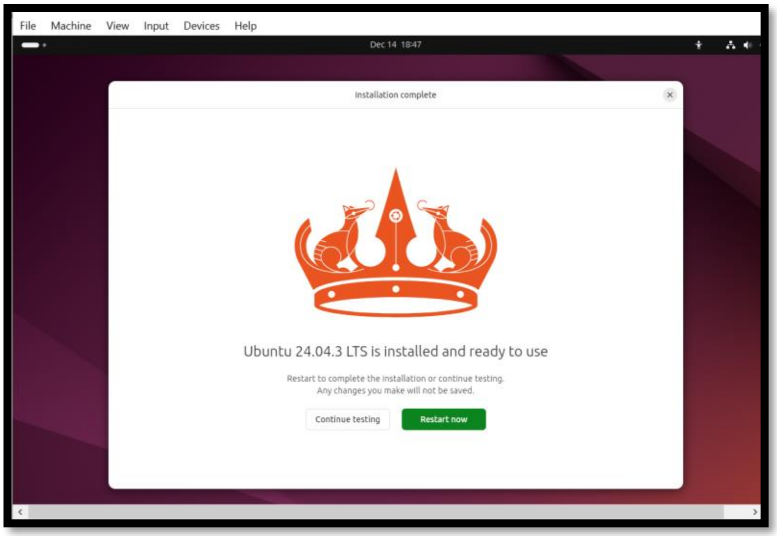

File Machine View Input Devices Help
Dec 14 12:47
Disk setup
How do you want to install Ubuntu?
Erase disk and install Ubuntu
Start from scratch on your selected disk.
Advanced features...
None selected
Manual installation
For advanced users seeking customized disk setups.
Back
Next

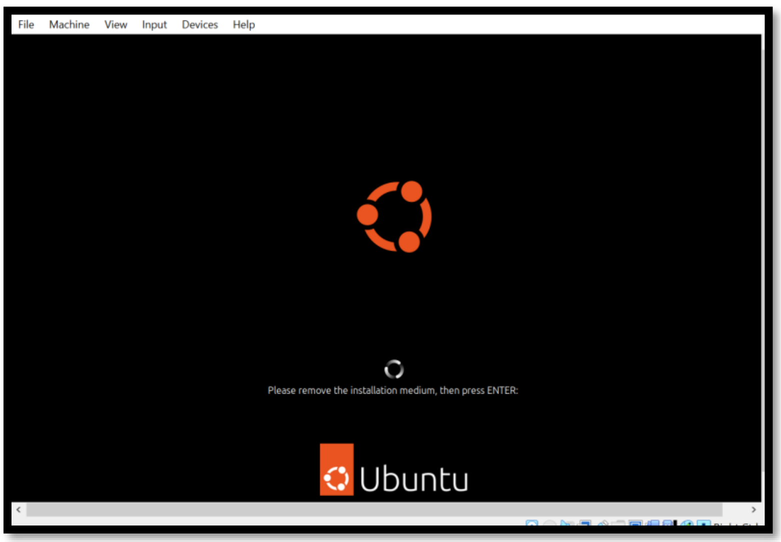

File Machine View Input Devices Help
Dec 14 12:48
Create your account
Create your account
Your name
sameen
Your computer's name
sameen-VirtualBox
Your username
sameen
Password
Show
Weak password
Confirm password
Require my password to log in
Use Active Directory
Back
Next

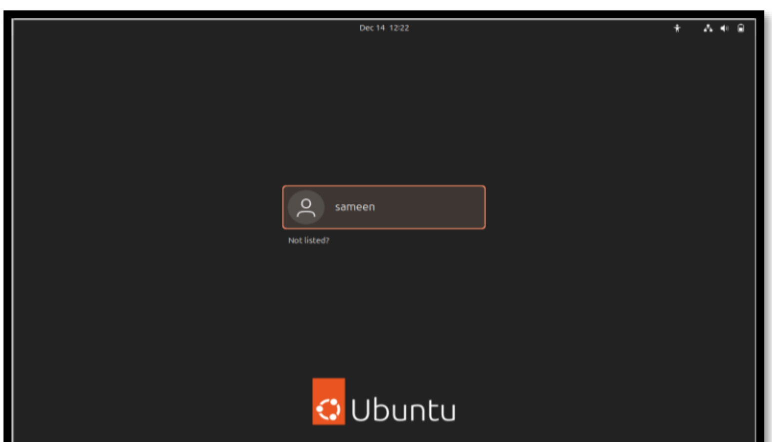

File Machine View Input Devices Help
Dec 14 12:49
Select your timezone
Location
Karachi (Sindh, Pakistan)
Timezone
Asia/Karachi
Back
Next

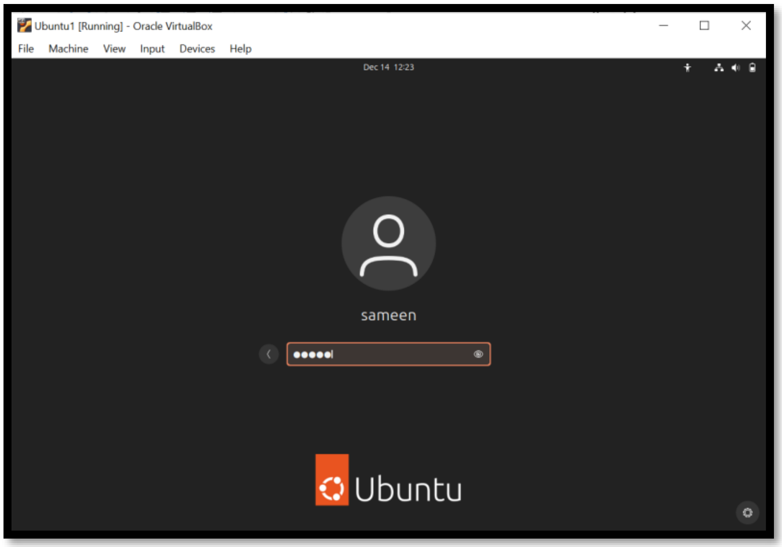

File Machine View Input Devices Help
Dec 14 17:50
Ready to install
Review your choices
General
Disk setup
Erase disk and install Ubuntu
Installation disk
VBOX HARDDISK sda
Applications
Default selection
Security & more
Disk encryption
None
Proprietary software
Drivers
Partitions
partition sda1 created
partition sda2 formatted as ext4 used for /
Back
Install

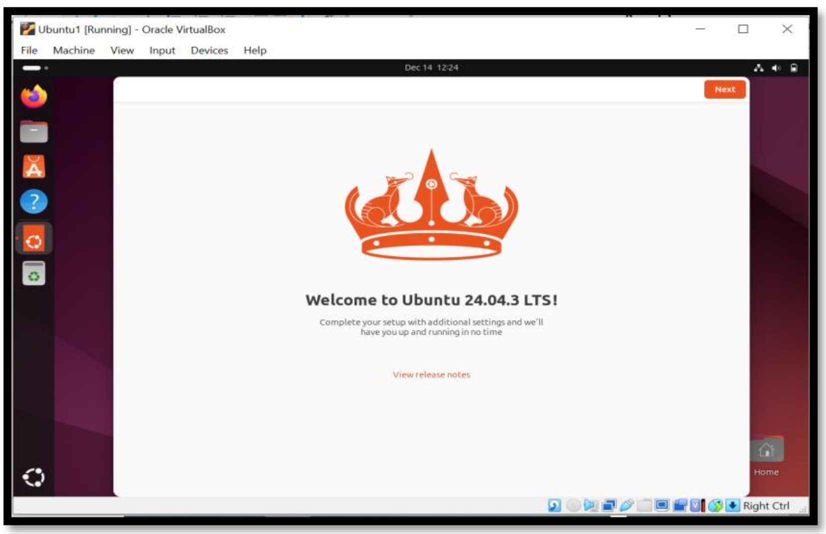

File Machine View Input Devices Help
Dec 14 17:51
Ubuntu 24.04.3 LTS
Fast, free and full of new features
The latest version makes computing easier than ever.
Whether you're a developer, creator, gamer or administrator you'll find new tools to improve your productivity and enhance your experience in this release.
Copying files...

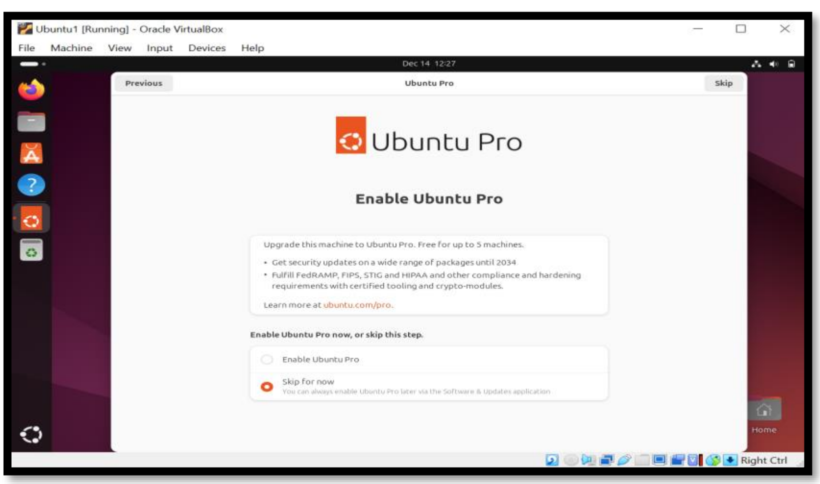

File Machine View Input Devices Help
Dec 14 18:47
Installation complete
Ubuntu 24.04.3 LTS is installed and ready to use
Restart to complete the installation or continue testing.
Any changes you make will not be saved.
Continue testing
Restart now

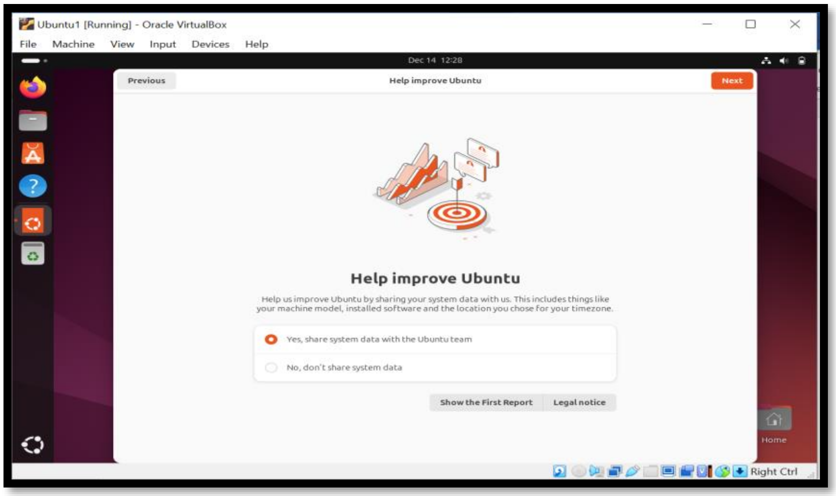

File Machine View Input Devices Help
Please remove the installation medium, then press ENTER:
Ubuntu

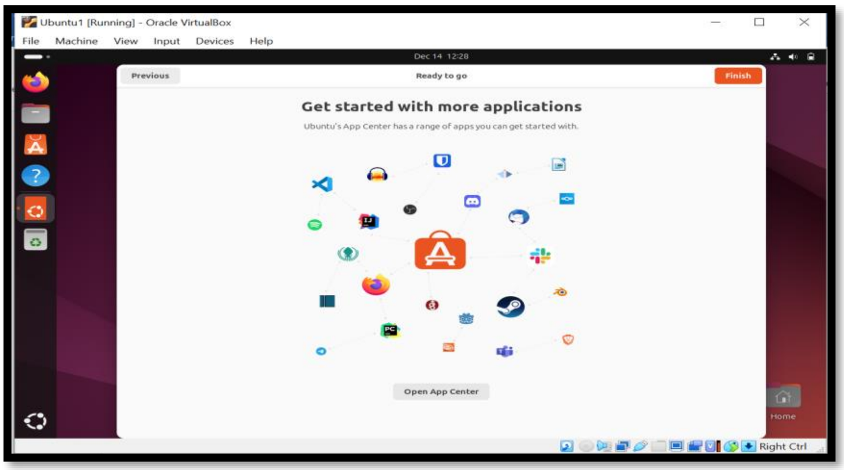

Dec 14 12:22
sameen
Not listed?
Ubuntu

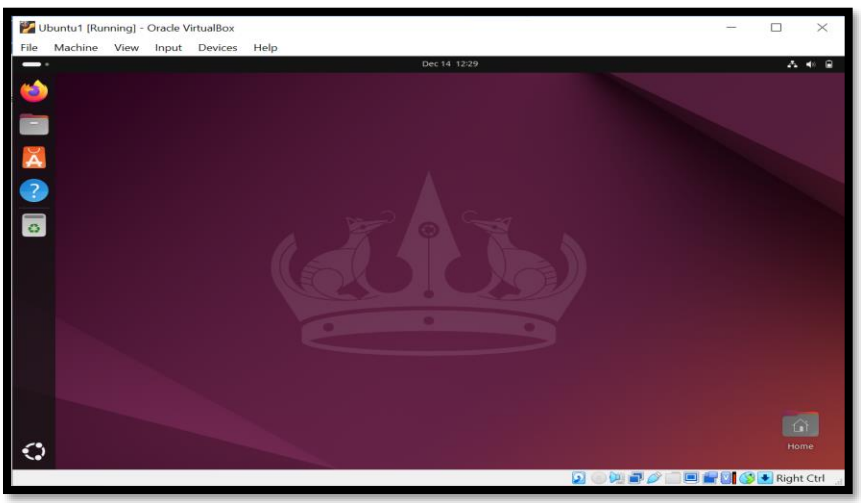

Ubuntu1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Dec 14 12:23
sameen
Ubuntu


Ubuntu1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Dec 14 12:24
Next
Welcome to Ubuntu 24.04.3 LTS!
Complete your setup with additional settings and we'll have you up and running in no time
View release notes
Home
Right Ctrl

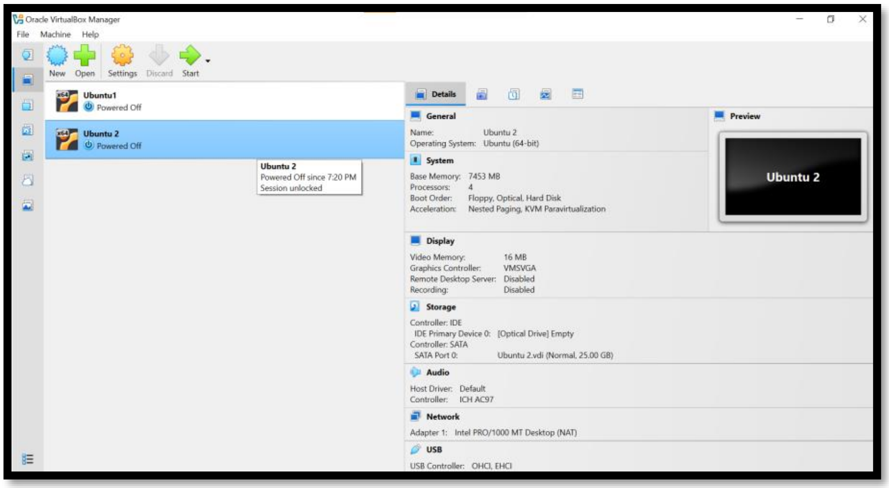

Ubuntu1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Dec 14 12:27
Previous
Ubuntu Pro
Skip
Ubuntu Pro
Enable Ubuntu Pro
Upgrade this machine to Ubuntu Pro. Free for up to 5 machines.
Get security updates on a wide range of packages until 2034
Fulfill FedRAMP, FIPS, STIG and HIPAA and other compliance and hardening requirements with certified tooling and crypto-modules.
Learn more at ubuntu.com/pro.
Enable Ubuntu Pro now, or skip this step.
Enable Ubuntu Pro
Skip for now
Home
Right Ctrl



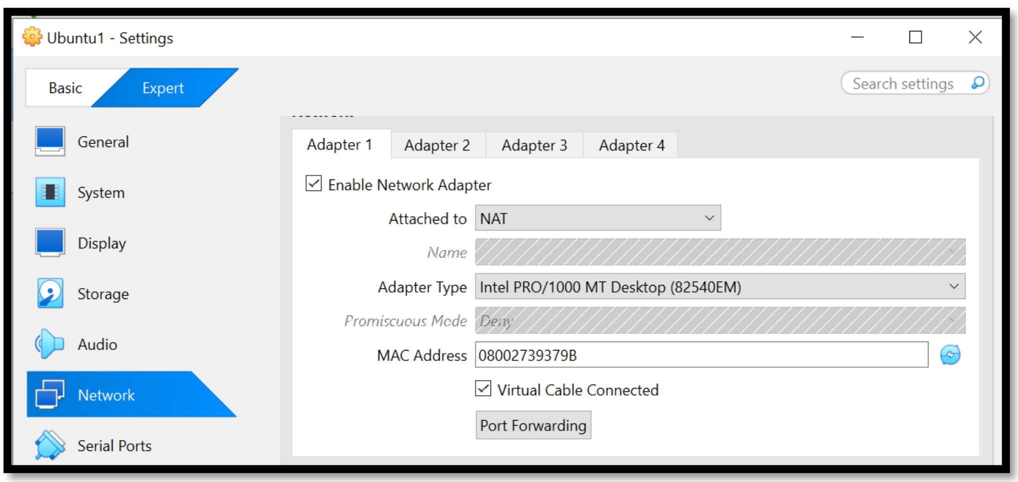


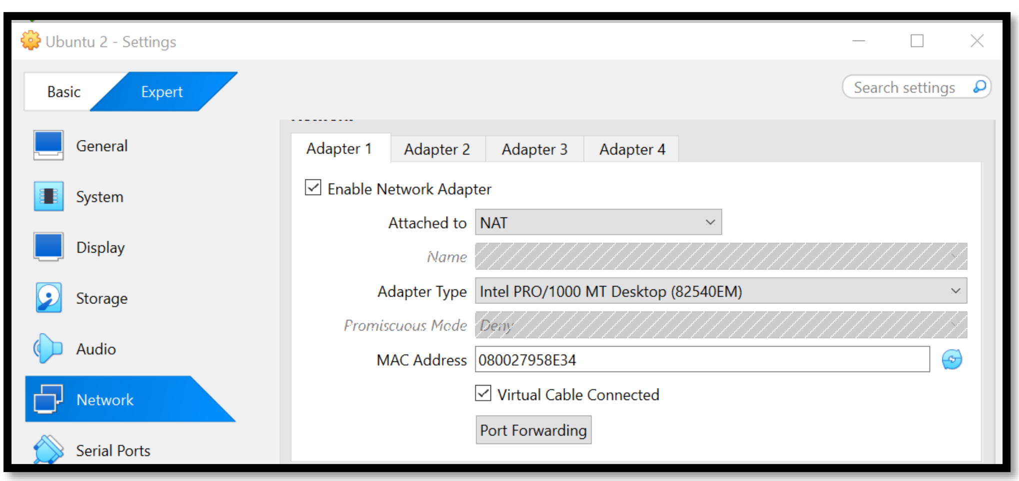


2. A second virtual machine (Ubuntu 2) was created using the same Ubuntu ISO file.



3. NAT networking mode was selected to allow communication between the two VMs.

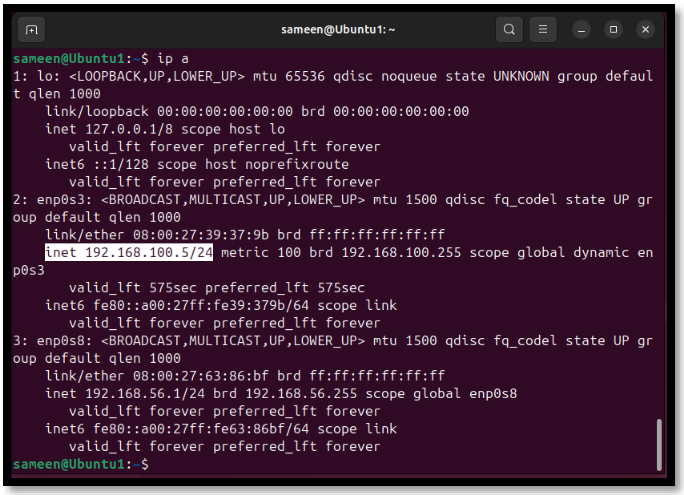




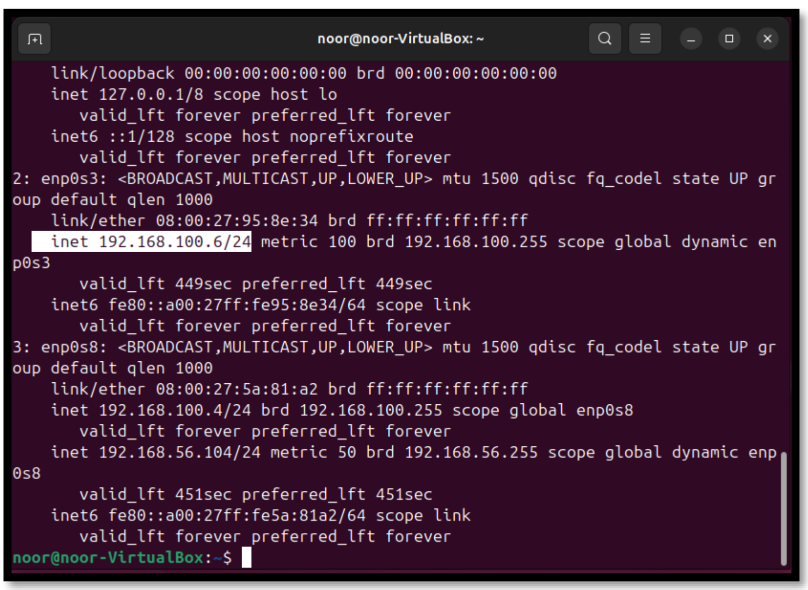
4. Both Ubuntu systems booted successfully.

Task 3 – Network Configuration and Ping Test

1. Both Ubuntu VMs obtained IP addresses in the same network range.
2. IP addresses were verified using the command: ip a
   - Ubuntu 1 (sameen user) IP: 192.168.100.5

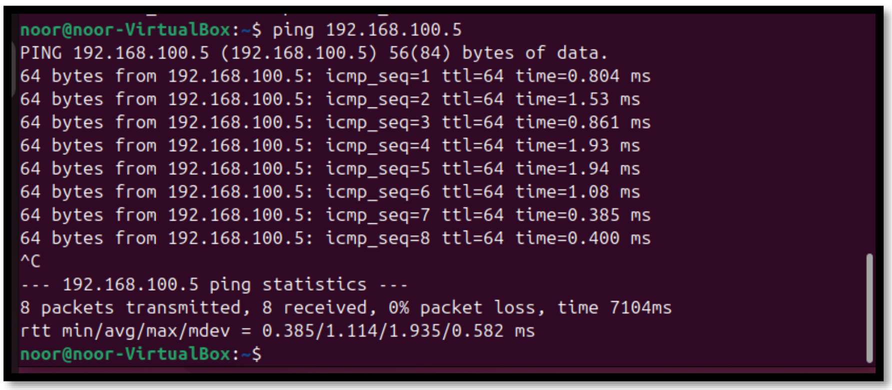


   - Ubuntu 2 (noor user) IP: 192.168.100.6



3. Ping test results:
   - From Ubuntu 1 to Ubuntu 2: ping 192.168.100.6

- From Ubuntu 2 to Ubuntu 1: ping 192.168.100.5



4. Successful replies confirmed that networking between both machines was correctly configured.

Server Configuration and Implementation

FTP SERVER

INTRODUCTION

File Transfer Protocol (FTP) is a network protocol used to transfer files between a client and a server over a TCP/IP network. It provides a simple way to upload and download files between systems. In this project, an FTP server is configured on one Ubuntu virtual machine using vsftpd, while another Ubuntu virtual machine is used as an FTP client to test file transfer functionality.

OBJECTIVE

The main objectives of this task are:

- To configure an FTP server on Ubuntu
- To allow secure access for local users
- To transfer files between two virtual machines

- To verify successful communication between client and server

SYSTEM SETUP

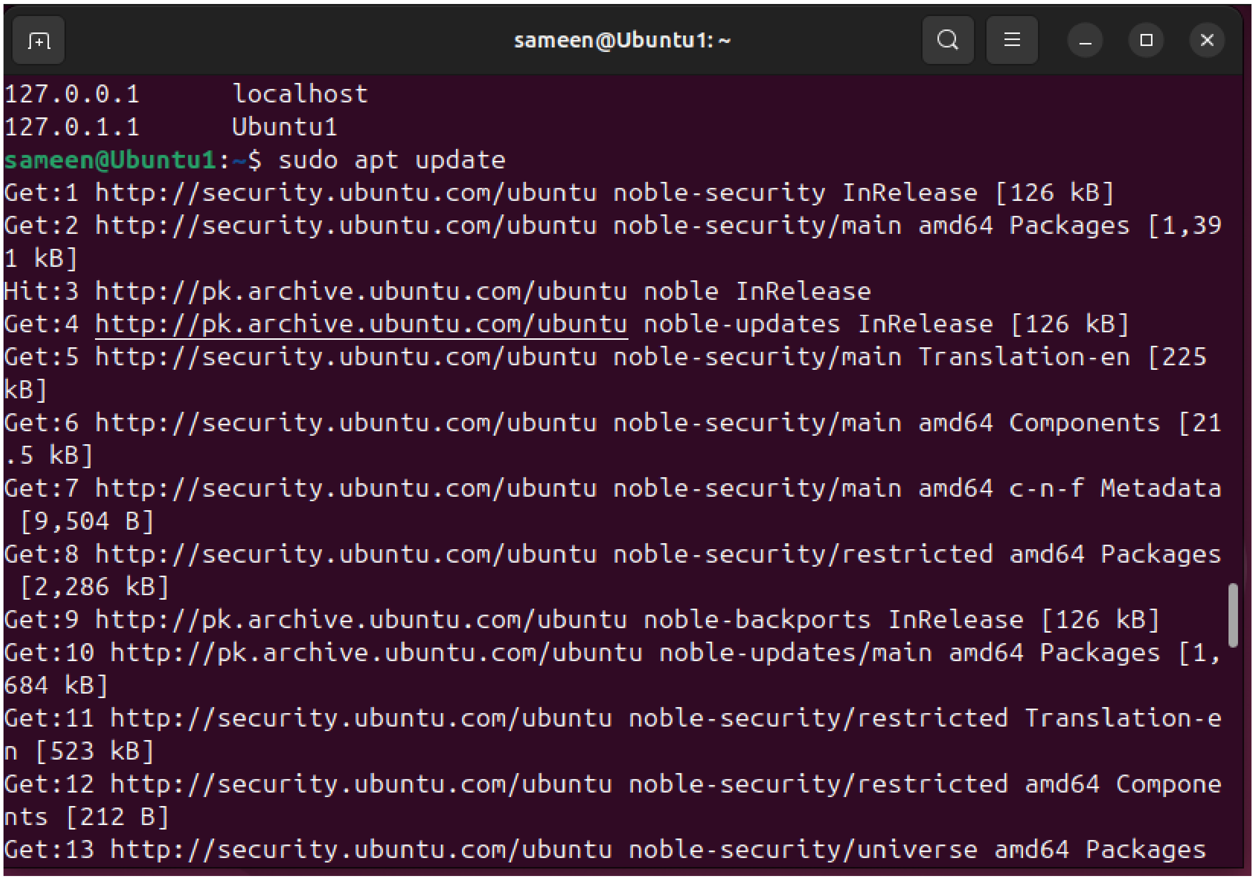
- **Operating System:** Ubuntu (VirtualBox)
- **Server Machine:** Ubuntu 1 (FTP Server)
- **Client Machine:** Ubuntu 2 (FTP Client)
- **FTP Software:** vsftpd (Very Secure FTP Daemon)

IMPLEMENTATION

Step 1: Update System Packages (Server Side)

The package list was updated to ensure the latest software versions are available.

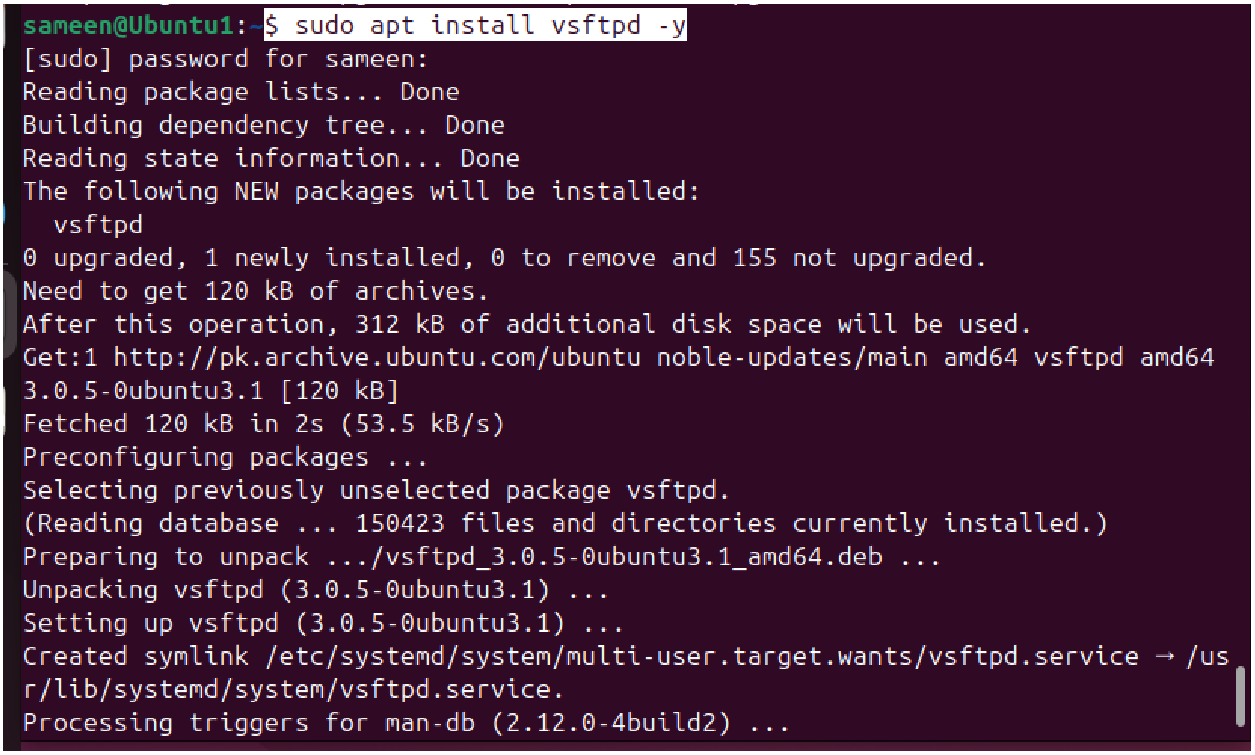
sudo apt update



Step 2: Install FTP Server (vsftpd)

The vsftpd package was installed on Ubuntu 1 to provide FTP services.

sudo apt install vsftpd -y



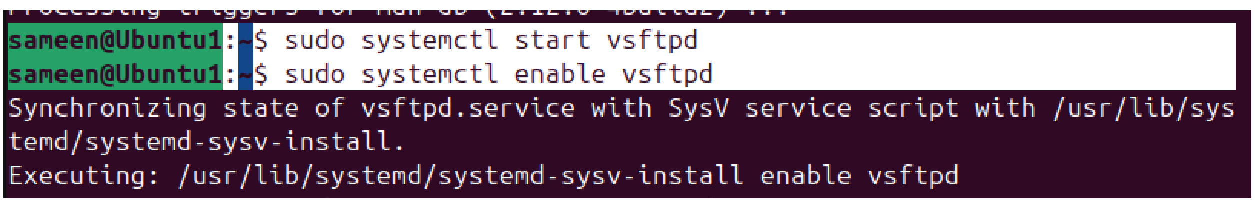
Step 3: Start and Enable FTP Service

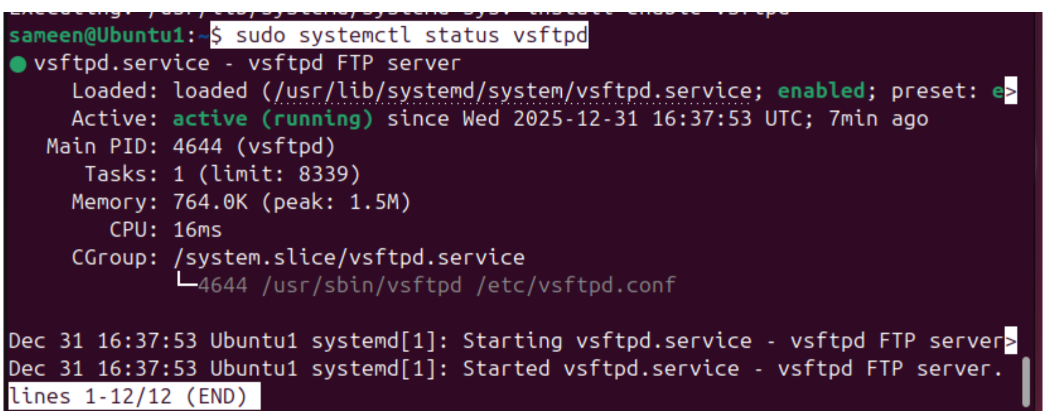
The FTP service was started and enabled to run automatically at system startup.

sudo systemctl start vsftpd

sudo systemctl enable vsftpd

sudo systemctl status vsftpd

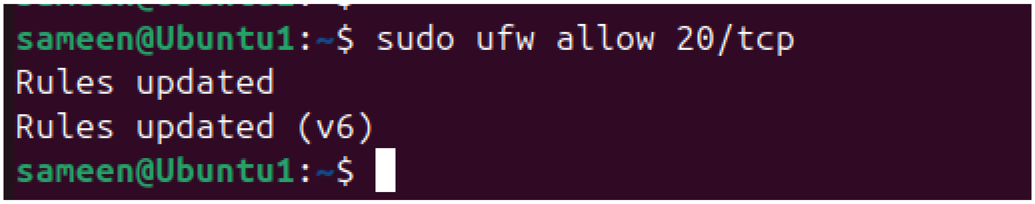




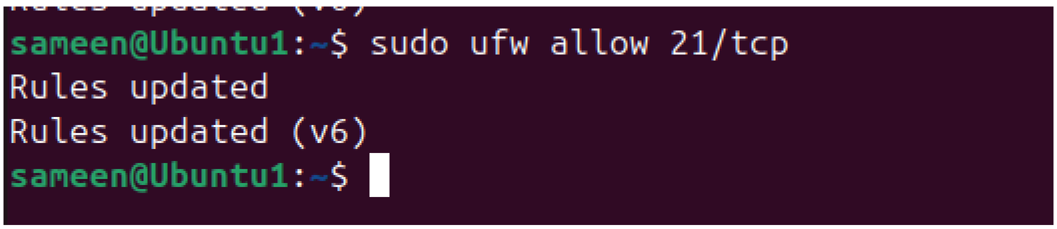
Step 4: Configure Firewall Rules

FTP ports were allowed through the firewall to enable communication.

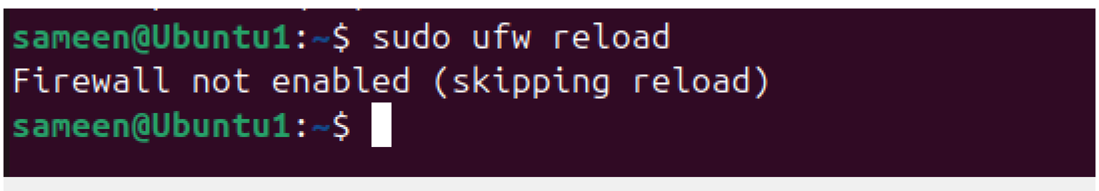
sudo ufw allow 20/tcp



sudo ufw allow 21/tcp



sudo ufw reload



Step 5: Configure FTP Settings

The main configuration file was edited to allow local users and file uploads.

sudo nano /etc/vsftpd.conf

The following settings were ensured or added:

local_enable=YES

write_enable=YES

chroot_local_user=YES

allow_writeable_chroot=YES

These settings allow local users to log in, upload files, and restrict them to their home directories for security.

Step 6: Restart FTP Service

After configuration changes, the FTP service was restarted.

sudo systemctl restart vsftpd

```
sameen@Ubuntu1:~$ sudo nano /etc/vsftpd.conf
sameen@Ubuntu1:~$ sudo systemctl restart vsftpd
```

Step 7: Create FTP User (Server Side)

A new user account was created for FTP access.

sudo adduser ftpuser

This user is used for FTP authentication and file transfers.

```
sameen@Ubuntu1:~$ sudo adduser ftpuser
info: Adding user `ftpuser' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `ftpuser' (1001) ...
info: Adding new user `ftpuser' (1001) with group `ftpuser (1001)' ...
info: Creating home directory `/home/ftpuser' ...
info: Copying files from `/etc/skel' ...
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
```

```
Changing the user information for ftpuser
Enter the new value, or press ENTER for the default
        Full Name []: Sameen Noor
        Room Number []: xx
        Work Phone []: 0315-xxxxxxx
        Home Phone []: 0315-xxxxxxx
        Other []: None
Is the information correct? [Y/n] Y
info: Adding new user `ftpuser' to supplemental / extra groups `users' ...
info: Adding user `ftpuser' to group `users' ...
sameen@Ubuntu1:~$
```

Step 8: Create Test File (Client Side)

On Ubuntu 2, a test file was created.

Note: The file name was intentionally changed to verify file transfer clearly.

echo "This is FTP testing file" > ftp_test.txt

```
noor@noor-VirtualBox:~$ echo "This is FTP testing file" > ftp_test.txt
```

Step 9: Connect to FTP Server from Client

The client connected to the FTP server using its IP address.

ftp 192.168.100.5

```
noor@noor-VirtualBox:~$ ftp 192.168.100.5
Connected to 192.168.100.5.
220 (vsFTPd 3.0.5)
```

```
Name (192.168.100.3:noor): ftpuser
331 Please specify the password.
Password:
```

Step 10: User Authentication

The FTP user credentials were entered:

Username: ftpuser

Password: (user-defined)

Successful login displayed the ftp> prompt.

```
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>
```

Step 11: Upload File to Server

The test file was uploaded from the client to the server.

put ftp_test.txt

A successful transfer message confirmed proper FTP operation.
RESULT:

FTP server was installed and configured successfully

Client connected to the server without errors

File transfer completed successfully

Changed file name was uploaded and verified on the server

```
ftp> put ftp_test.txt
local: ftp_test.txt remote: ftp_test.txt
229 Entering Extended Passive Mode (|||53110|)
150 Ok to send data.
100% |***********************************|    25       71.59 KiB/s    00:00 ETA
226 Transfer complete.
25 bytes sent in 00:00 (9.11 KiB/s)
ftp>
```

CONCLUSION

The FTP server was successfully configured on Ubuntu using vsftpd. The client machine connected to the server using valid credentials and transferred a file without any issues. This project demonstrates effective FTP server setup, secure user access, and reliable file transfer between virtual machines in a controlled network environment.

WEB SERVER

Introduction

A web server is software that hosts and delivers web content to users over a network. In this phase, the **Apache2** web server was installed and configured on an Ubuntu-based virtual machine. Apache is a widely-used, open-source server solution known for its reliability and flexibility. The setup was tested both locally and from a remote client machine to ensure proper functionality.

Role Distribution

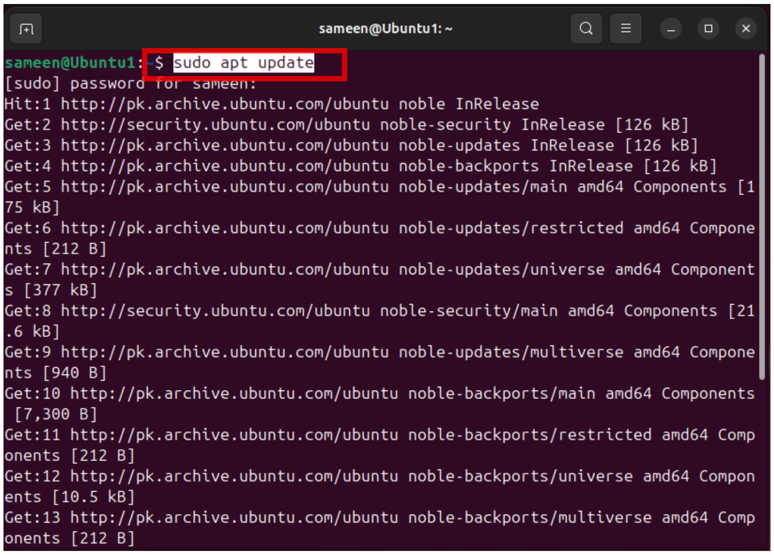
- **Ubuntu 1:** Server operating system running Apache.
- **Ubuntu 2:** Client operating system for remote testing.
- **Web Server Software:** Apache2 (latest stable version).

Step 1: System Update

The first step involved updating the system's package repository to ensure access to the latest software versions.

sudo apt update

Purpose: Refreshes the local package index to synchronize with the Ubuntu repositories.

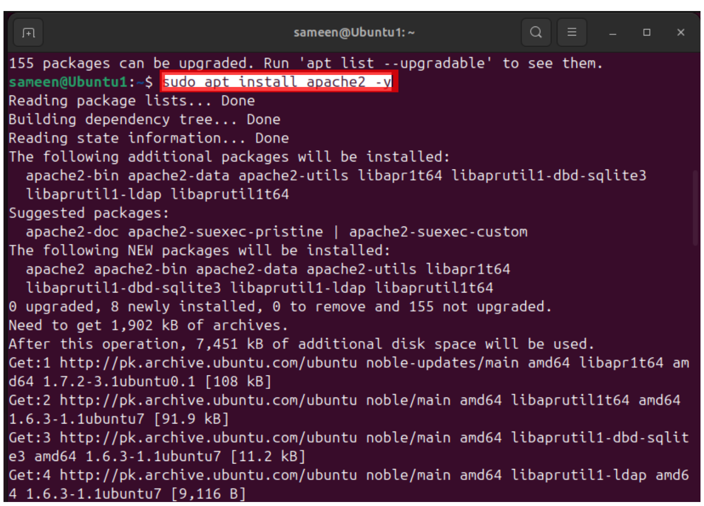


Step 2: Apache Installation

Apache2 was installed using the default package manager.

sudo apt install apache2 -y

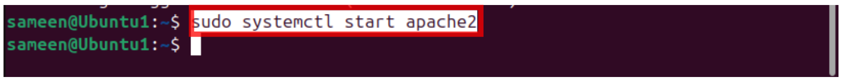
Purpose: Installs the Apache web server with automatic confirmation (`-y` flag).



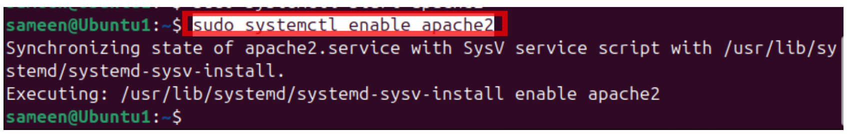
Step 3: Starting and Enabling Apache Service

After installation, the Apache service was started, enabled for automatic startup on boot, and its status verified.

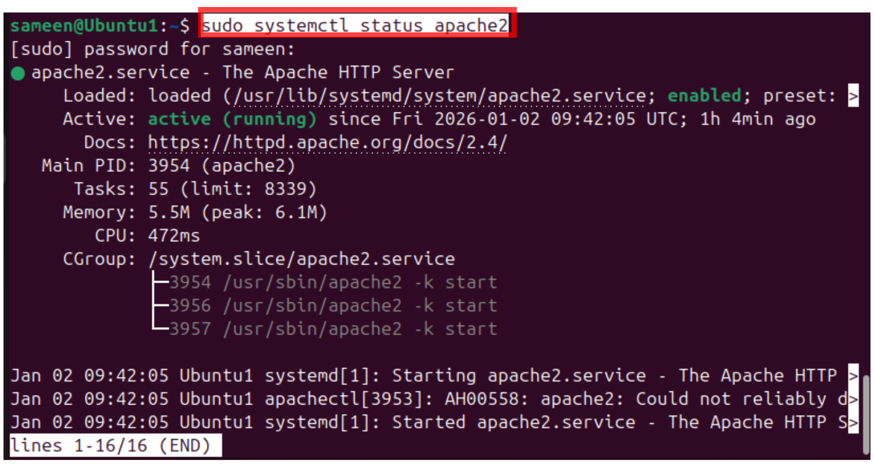
sudo systemctl start apache2



sudo systemctl enable apache2



sudo systemctl status apache2



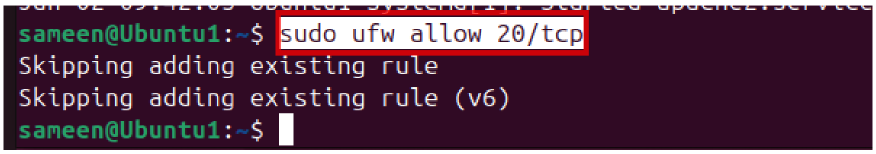
Purpose:

- `start` → Immediately launches the Apache service.
- `enable` → Configures Apache to launch automatically at system startup.
- `status` → Confirms that Apache is running without errors.

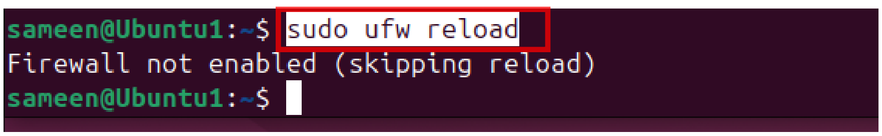
Step 4: Firewall Configuration

To allow external HTTP access, Apache traffic was permitted through the system firewall.

sudo ufw allow 'Apache'



sudo ufw reload



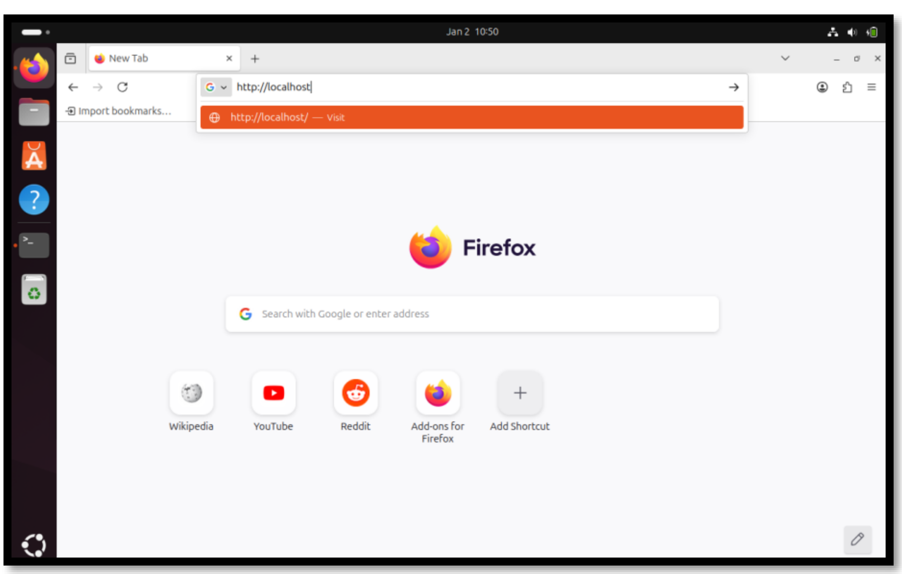
Explanation:

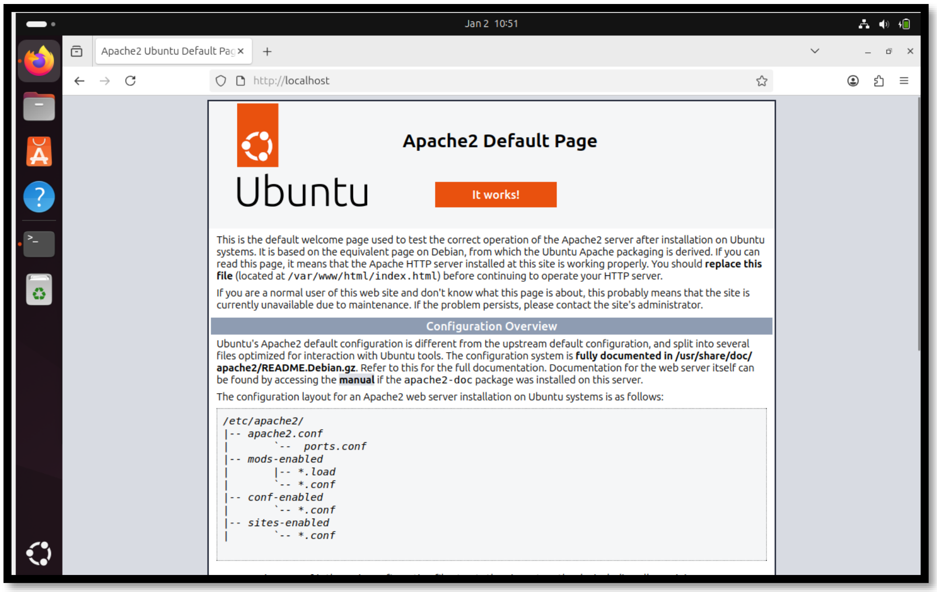
These commands allow HTTP traffic (port 80) so that the web server can be accessed over the network.

Step 5: Local Verification

The web server was tested locally on the server machine (Ubuntu 1) by accessing:

URL: `http://localhost`

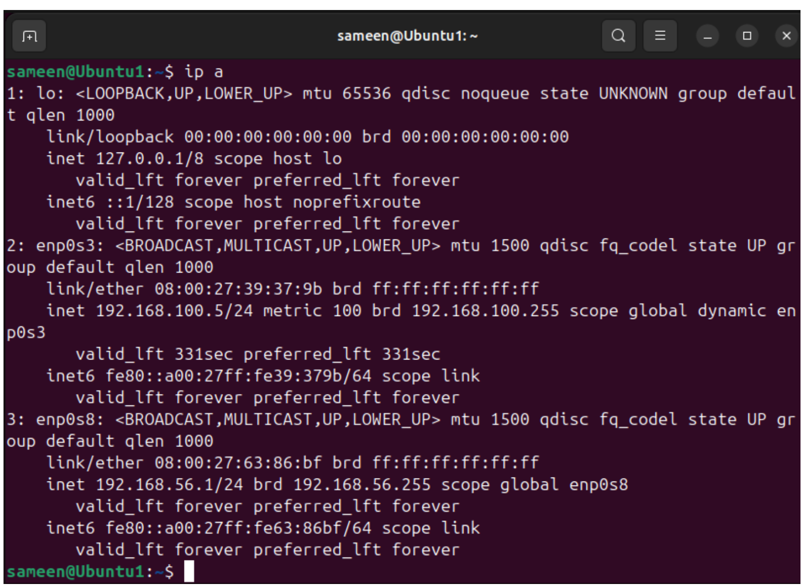




Result: The default **Apache2 Ubuntu Default Page** appeared, confirming that the server was operational locally.

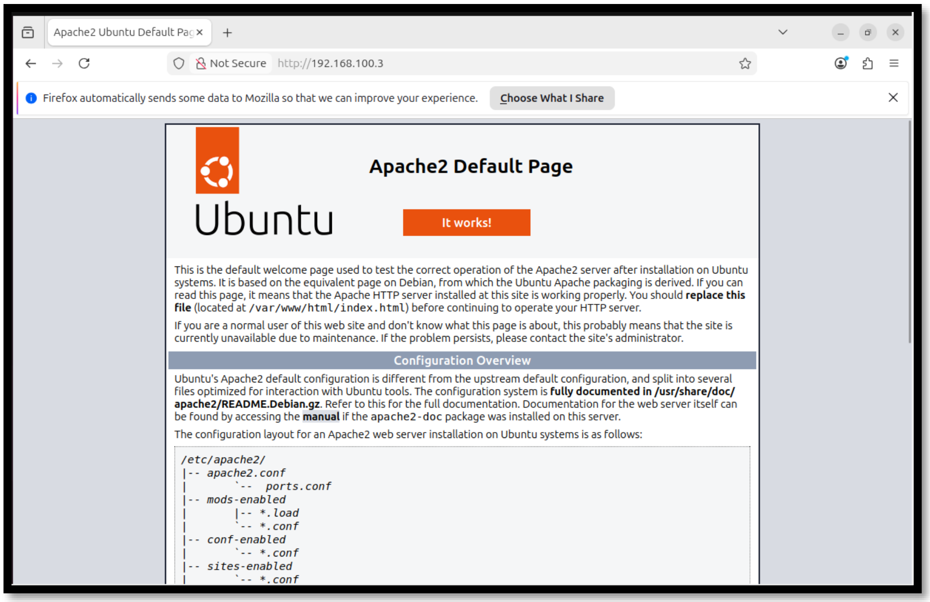
Step 6 – Test Web Server Apache from Client Machine (Ubuntu 2)

The IP address of the server machine (Ubuntu 1) was obtained using:



The web server was then accessed from the second Ubuntu virtual machine using:

http://192.168.100.5

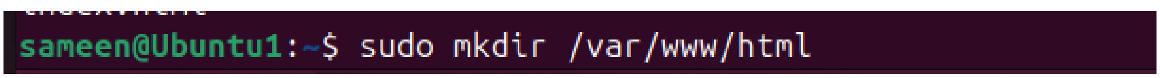


Step 7 – Create a Custom Web Page

Create Directory
Run the following command in the terminal:

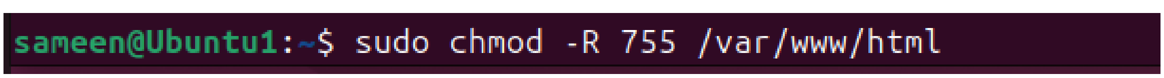
sudo mkdir /var/www/html

The folder has now been created.



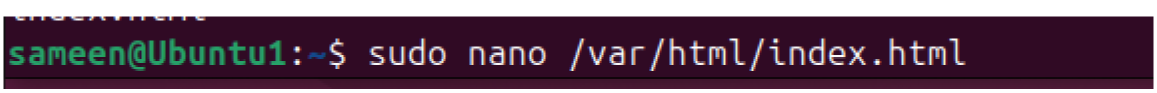
Set Permissions
Assign the correct permissions to the directory to ensure Apache can access its contents:

sudo chmod -R 755 /var/www/html

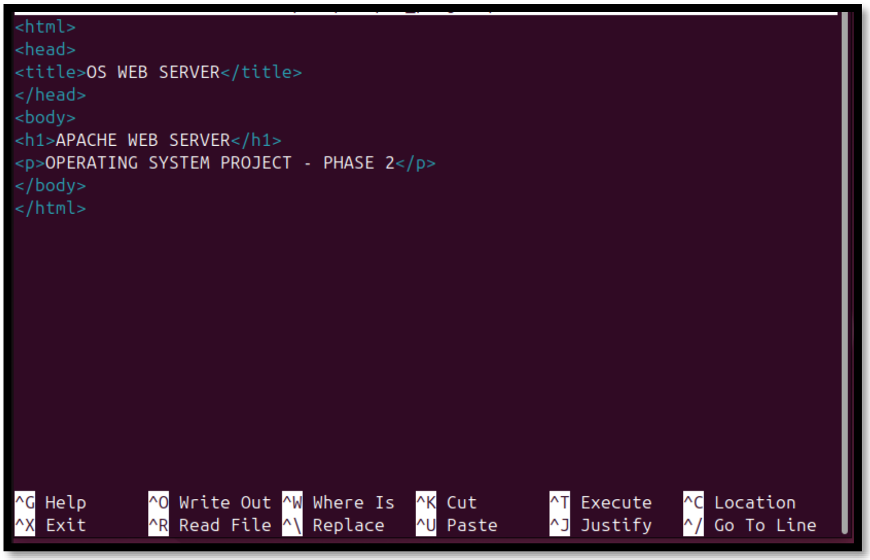


Open the File Again
Now open your HTML file in the html directory using the text editor:

sudo nano /var/www/html/index.html

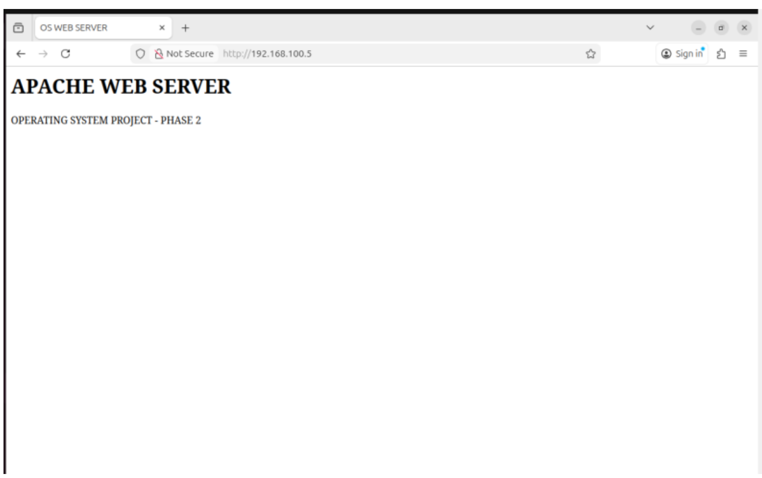


Write the code in index.html.



CTRL + O → ENTER → CTRL + X

Note: Refresh browser on Ubuntu 2.



Conclusion:
In this task, the web server on Ubuntu was properly set up and configured. Files were successfully created, modified, and renamed within the server's directory, and all changes were verified by accessing the pages through a web browser. This demonstrates effective server configuration, proper file management, and successful interaction between the server and client interface.

DNS SERVER

Introduction

Domain Name System (DNS) is a critical network service that translates human-readable domain names into IP addresses. Without DNS, users would need to remember IP addresses to access network services. In this phase of the Operating Systems project, a DNS server is configured on Ubuntu Linux to provide name resolution services within a local network.

This report documents the complete DNS server setup using BIND9 on Ubuntu, including server and client configuration. All steps are explained clearly with reference to screenshots taken during implementation.

Objectives

- To install and configure a DNS server on Ubuntu using BIND9
- To understand DNS zone creation and configuration
- To configure a DNS client to use the DNS server
- To verify DNS name resolution within a local network

Role of Machines

Ubuntu-1 (DNS Server)

- Acts as the primary DNS server
- Runs BIND9 service
- Stores forward and reverse zone files
- Resolves domain names for client machines

Ubuntu-2 (DNS Client)

- Acts as a DNS client
- Sends DNS queries to the DNS server
- Tests name resolution using ping and other commands

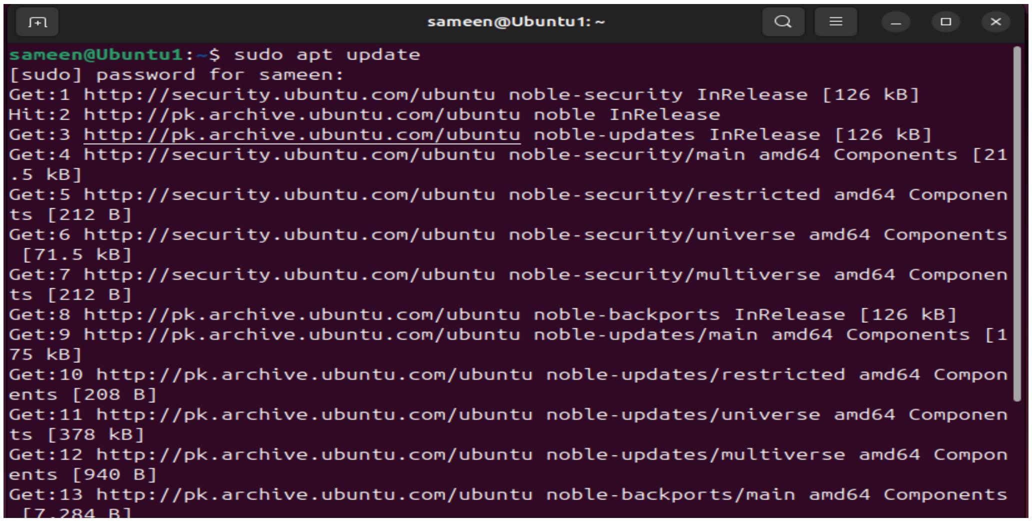
Network Configuration

- DNS Server IP Address: `192.168.100.5`
- DNS Client IP Address: `192.168.100.6`

Step 1 – Install DNS Server (BIND9) - Ubuntu 1

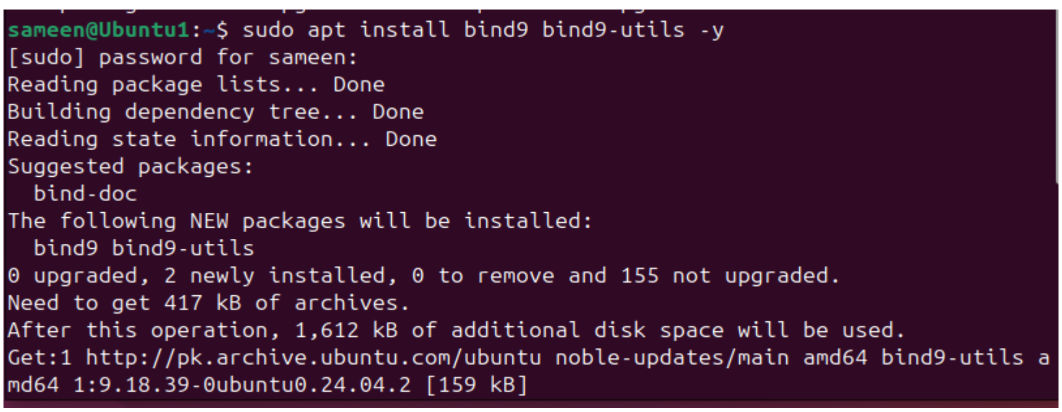
BIND9 is the most commonly used DNS server software on Linux systems. It translates domain names into IP addresses.

`sudo apt update`



```
sameen@Ubuntu1:~$ sudo apt update
[sudo] password for sameen:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://pk.archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://pk.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21
.5 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Componen
ts [212 B]
Get:6 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components
 [71.5 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Componen
ts [212 B]
Get:8 http://pk.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:9 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [1
75 kB]
Get:10 http://pk.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Compon
ents [208 B]
Get:11 http://pk.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Componen
ts [378 kB]
Get:12 http://pk.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Compon
ents [940 B]
Get:13 http://pk.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components
 [7,284 B]
```

`sudo apt install bind9 bind9utils bind9-doc -y`

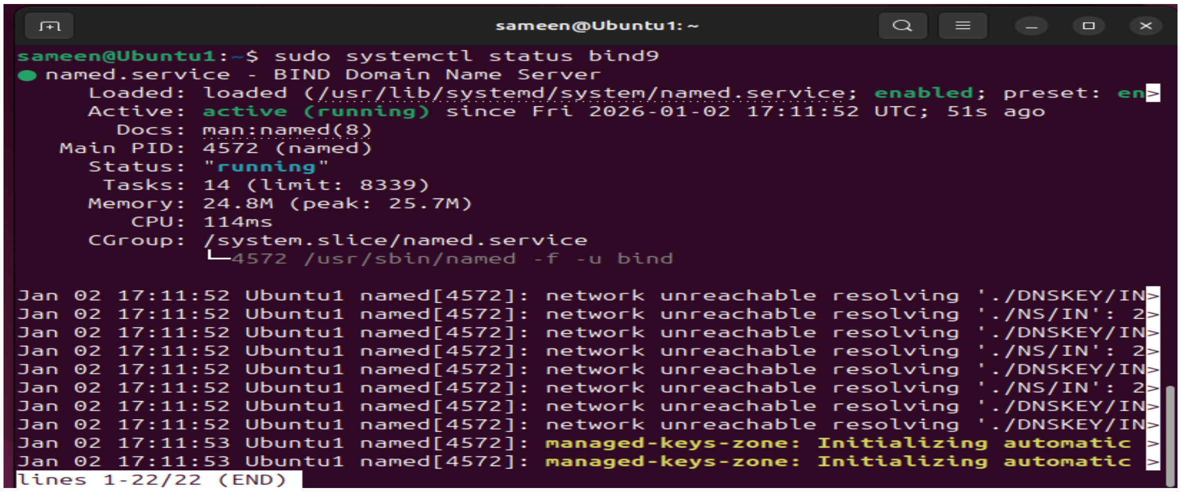


```
sameen@Ubuntu1:~$ sudo apt install bind9 bind9-utils -y
[sudo] password for sameen:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  bind-doc
The following NEW packages will be installed:
  bind9 bind9-utils
0 upgraded, 2 newly installed, 0 to remove and 155 not upgraded.
Need to get 417 kB of archives.
After this operation, 1,612 kB of additional disk space will be used.
Get:1 http://pk.archive.ubuntu.com/ubuntu noble-updates/main amd64 bind9-utils a
md64 1:9.18.39-0ubuntu0.24.04.2 [159 kB]
```

Step 2 – Check DNS Service Status (Verify BIND is Running)

`sudo systemctl status bind9`

Service showing **active (running)**

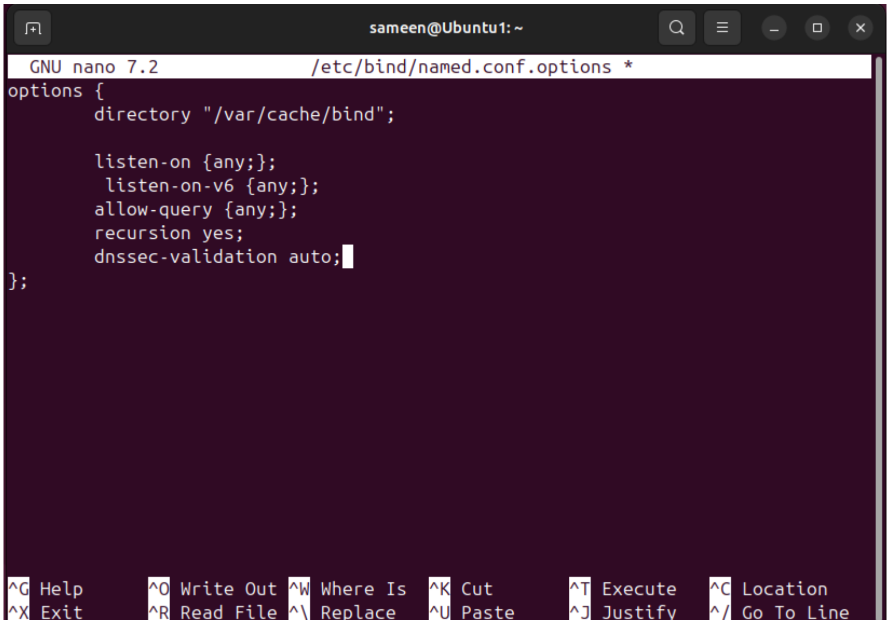


Step 3 – Configure DNS Options (Ubuntu-1)

sudo nano /etc/bind/named.conf.options

Replace **everything inside** with:

```
options {
    directory "/var/cache/bind";
    listen-on { any; };
    listen-on-v6 { any; };
    allow-query { any; };
    recursion yes;
    dnssec-validation auto;
};
```



Step 4 – Define DNS Zone (Ubuntu-1)

sudo nano /etc/bind/named.conf.local

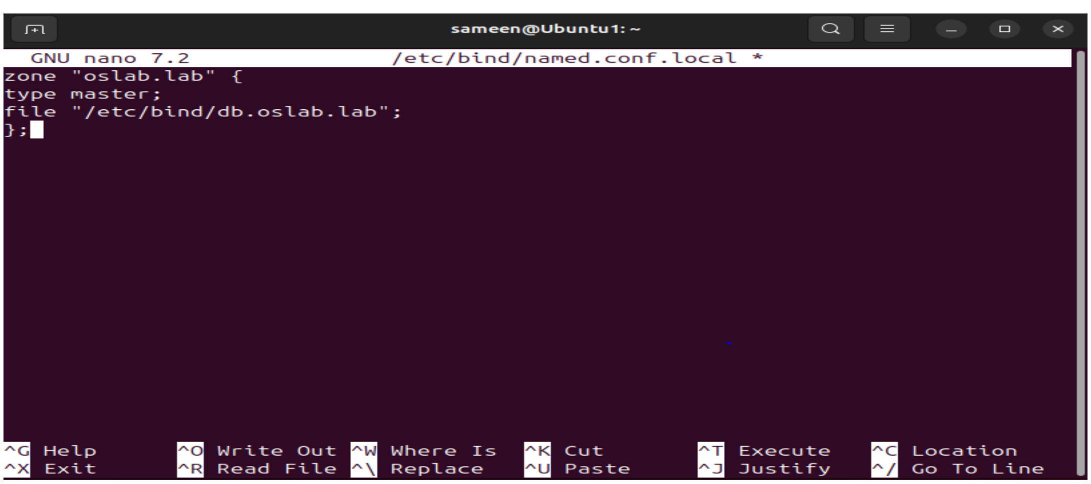
Add:

zone "oslab.lab" {

type master;

file "/etc/bind/db.oslab.lab";

};



Step 5 – Create Zone File (Ubunt-1)

sudo cp /etc/bind/db.local /etc/bind/db.oslab.lab

```
sameen@Ubuntu1:~$ sudo cp /etc/bind/db.local /etc/bind/db.oslab.lab
```

sudo nano /etc/bind/db.oslab.lab

```
sameen@Ubuntu1:~$ sudo nano /etc/bind/db.oslab.lab
```

Replace content with:

```
$TTL    604800
@       IN      SOA     oslab.lab. root.oslab.lab. (
                              2         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      oslab.lab.
@       IN      A       192.168.100.5
www     IN      A       192.168.100.5
```

```
sameen@Ubuntu1: ~
 GNU nano 7.2                    /etc/bind/db.oslab.lab *
;
; BIND data file for local loopback interface
;
$TTL    604800
@       IN      SOA     localhost. root.localhost. (
                              2         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      localhost.
@       IN      A       192.168.100.5
www     IN      A       192.168.100.5

^G Help       ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit       ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line
```

Step 6 – Check DNS Configuration (Ubuntu-1)

sudo named-checkconf

```
sameen@Ubuntu1:~$ sudo named-checkconf
sameen@Ubuntu1:~$
```

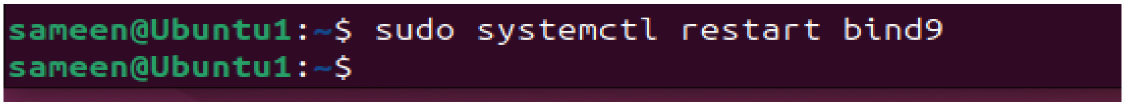
sudo named-checkzone oslab.lab /etc/bind/db.oslab.lab

Expected: OK

```
sameen@Ubuntu1:~$ sudo named-checkzone oslab.lab /etc/bind/db.oslab.lab
zone oslab.lab/IN: loaded serial 2
OK
sameen@Ubuntu1:~$
```

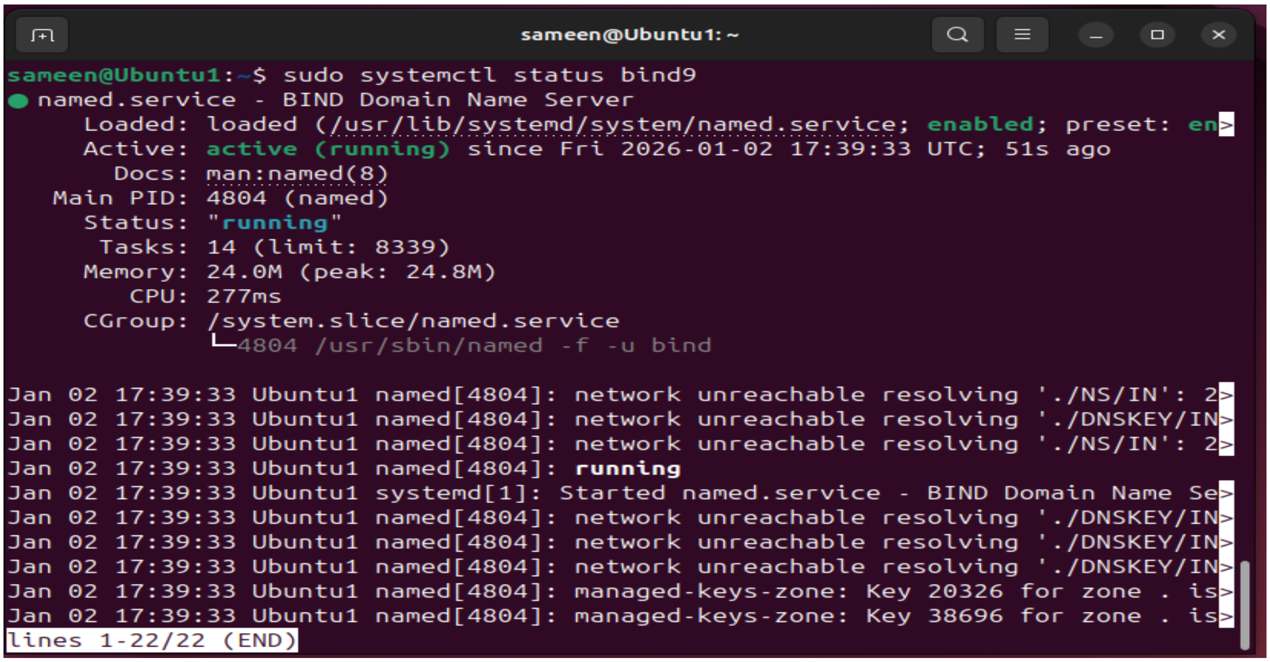
Step 7 – Restart DNS Server

sudo systemctl restart bind9



Verify:

`sudo systemctl status bind9`



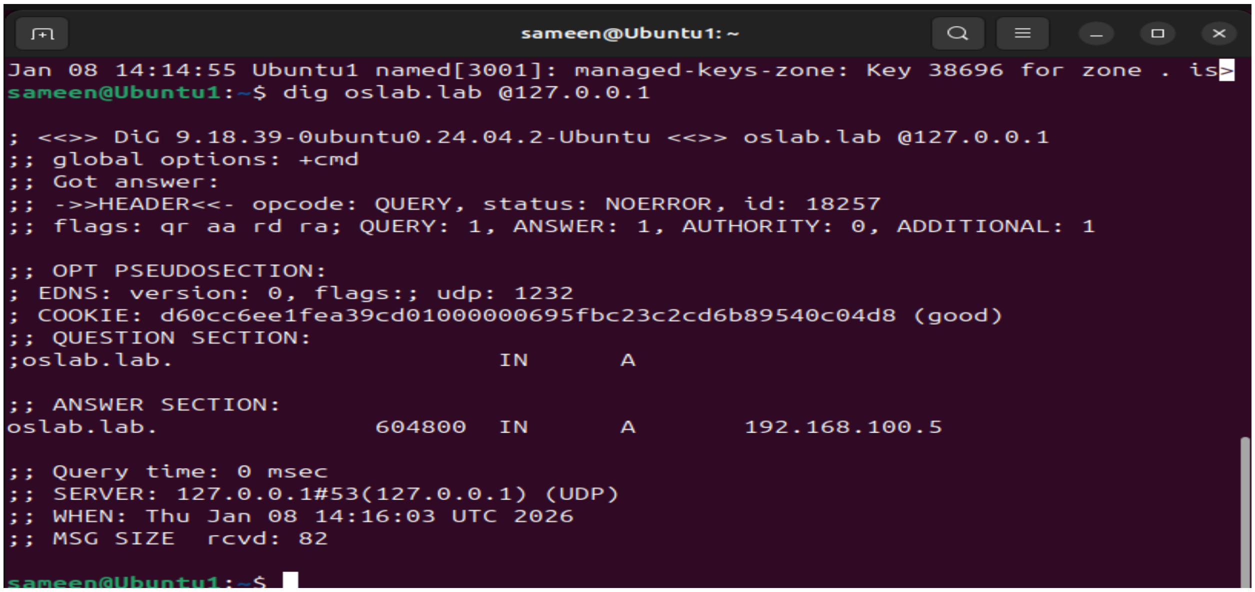
Step 8 – Test DNS Locally (Ubunt-1)

`dig oslab.lab @127.0.0.1`

Expected:

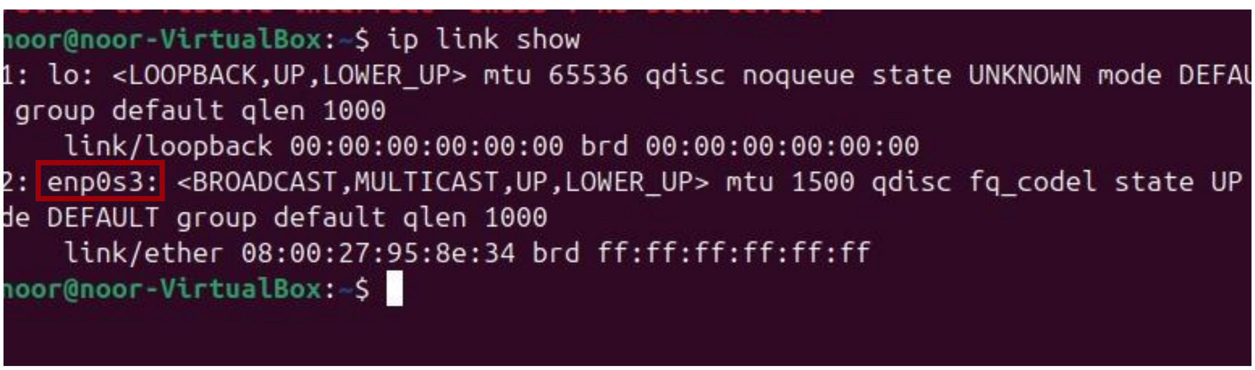
`ANSWER SECTION:`

`oslab.lab.    IN    A    192.168.100.5`



Check Available Network Interfaces
To identify the correct network interface name, use:

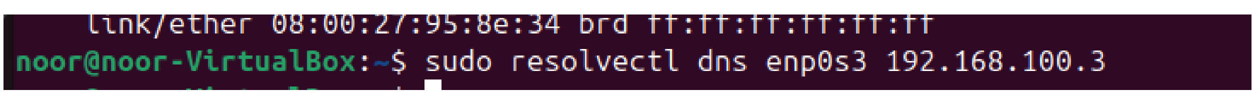
ip link show



Step 9 – Configure DNS Client (Ubuntu-2)

Set DNS server:

sudo resolvectl dns enp0s3 192.168.100.3



Set domain:

sudo resolvectl domain ens33 oslab.lab

```
noor@noor-VirtualBox:~$ sudo resolvectl dns enp0s3 192.168.100.3
noor@noor-VirtualBox:~$ sudo resolvectl domain enp0s3 oslab.lab
```

Verify:

resolvectl status

```
noor@noor-VirtualBox:~$ resolvectl status
```

```
noor@noor-VirtualBox:~$ resolvectl status
Global
         Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
  resolv.conf mode: stub

Link 2 (enp0s3)
    Current Scopes: DNS
         Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
Current DNS Server: 192.168.100.3
       DNS Servers: 192.168.100.3
        DNS Domain: oslab.lab
noor@noor-VirtualBox:~$
```

Step 10 – FINAL TEST (Ubuntu-2)

ping oslab.lab

```
noor@noor-VirtualBox:~$ ping oslab.lab
PING oslab.lab (192.168.100.3) 56(84) bytes of data.
64 bytes from 192.168.100.3: icmp_seq=1 ttl=64 time=0.409 ms
64 bytes from 192.168.100.3: icmp_seq=2 ttl=64 time=0.960 ms
64 bytes from 192.168.100.3: icmp_seq=3 ttl=64 time=1.42 ms
64 bytes from 192.168.100.3: icmp_seq=4 ttl=64 time=1.33 ms
64 bytes from 192.168.100.3: icmp_seq=5 ttl=64 time=0.401 ms
64 bytes from 192.168.100.3: icmp_seq=6 ttl=64 time=0.456 ms
```

Ping www.oslab.lab

```
noor@noor-VirtualBox:~$ ping www.oslab.lab
PING www.oslab.lab (192.168.100.3) 56(84) bytes of data.
64 bytes from 192.168.100.3: icmp_seq=1 ttl=64 time=0.808 ms
64 bytes from 192.168.100.3: icmp_seq=2 ttl=64 time=1.22 ms
64 bytes from 192.168.100.3: icmp_seq=3 ttl=64 time=0.862 ms
64 bytes from 192.168.100.3: icmp_seq=4 ttl=64 time=5.25 ms
64 bytes from 192.168.100.3: icmp_seq=5 ttl=64 time=1.22 ms
64 bytes from 192.168.100.3: icmp_seq=6 ttl=64 time=2.07 ms
64 bytes from 192.168.100.3: icmp_seq=7 ttl=64 time=0.734 ms
64 bytes from 192.168.100.3: icmp_seq=8 ttl=64 time=1.04 ms
64 bytes from 192.168.100.3: icmp_seq=9 ttl=64 time=1.06 ms
^C
--- www.oslab.lab ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8333ms
rtt min/avg/max/mdev = 0.734/1.583/5.247/1.347 ms
noor@noor-VirtualBox:~$
```

CONCLUSION

The DNS server was successfully configured on Ubuntu using BIND9. The server correctly resolved domain names to IP addresses for client systems, demonstrating proper DNS functionality. This implementation confirms effective name resolution within the network environment.

DHCP SERVER

1. Introduction

Dynamic Host Configuration Protocol (DHCP) is a network management protocol used to automatically assign IP addresses and other network configuration parameters to devices on a network. DHCP eliminates the need for manual IP configuration and ensures efficient and error-free network communication.

In this project, a DHCP server is installed and configured on **Ubuntu 1**, and a client machine **Ubuntu 2** is configured to obtain an IP address dynamically from the server.

2. Objective

The main objectives of this project are:

- To install and configure a DHCP server on Ubuntu
- To assign IP addresses dynamically to client machines
- To verify successful communication between server and client
- To understand DHCP working in a virtualized network environment

4. Role of Each Machine

Ubuntu 1 (Server)

- Acts as the **DHCP Server**
- Assigns IP addresses to clients
- Provides gateway and DNS information

Ubuntu 2 (Client)

- Acts as a **DHCP Client**
- Requests IP address from the DHCP server
- Uses **the** assigned IP for network communication

5. Network Configuration Used

Each virtual machine uses **two network adapters**:

1. NAT Adapter

- Provides internet access
- Automatically assigns IP via VirtualBox NAT

2. Host-Only Adapter

- Used for communication between server and client
- DHCP server operates on this network

Network Details:

- Host-Only Network: 192.168.56.0/24
- Server IP: 192.168.56.1

6. DHCP Server Installation (Ubuntu 1)

sudo apt update

This command updates the local package index to ensure the system installs the latest available versions of software packages.

```
       valid_lft forever preferred_lft forever
sameen@Ubuntu1:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://pk.archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://pk.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:4 http://pk.archive.ubuntu.com/ubuntu noble-backports InRelease
Fetched 126 kB in 8s (15.6 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
155 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

sudo apt install isc-dhcp-server

This command installs the ISC DHCP Server package, which provides the DHCP service required to assign IP addresses dynamically to clients.

```
sameen@Ubuntu1:~$ sudo apt install isc-dhcp-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  isc-dhcp-common
```

7. DHCP Server Configuration (Ubuntu 1)

sudo nano /etc/default/isc-dhcp-server

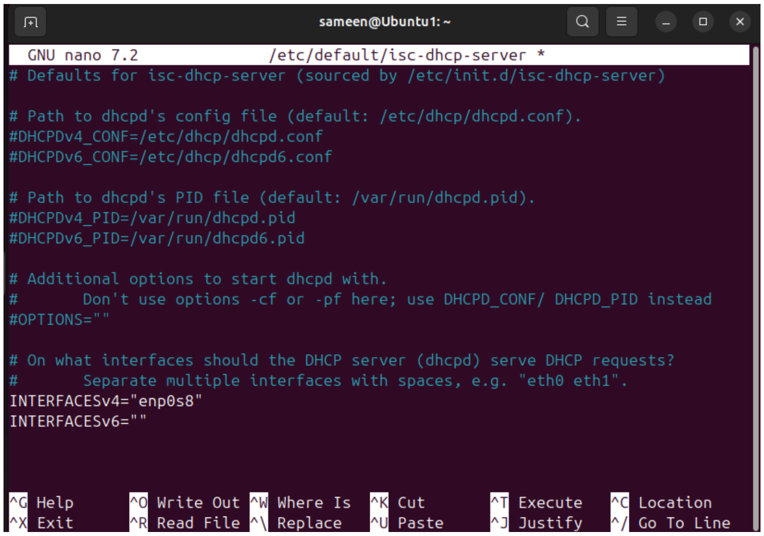
This command opens the DHCP default configuration file to specify which network interface the DHCP server should listen on.

```
sameen@Ubuntu1:~$ sudo nano /etc/dhcp/dhcpd.conf
```

Inside file, type :

INTERFACESv4="enp0s8"

This line binds the DHCP server to the Host-Only network interface, ensuring that IP addresses are assigned only on the internal network.



sudo nano /etc/dhcp/dhcpd.conf

This command opens the main DHCP configuration file where IP address ranges, subnet, gateway, and DNS settings are defined.

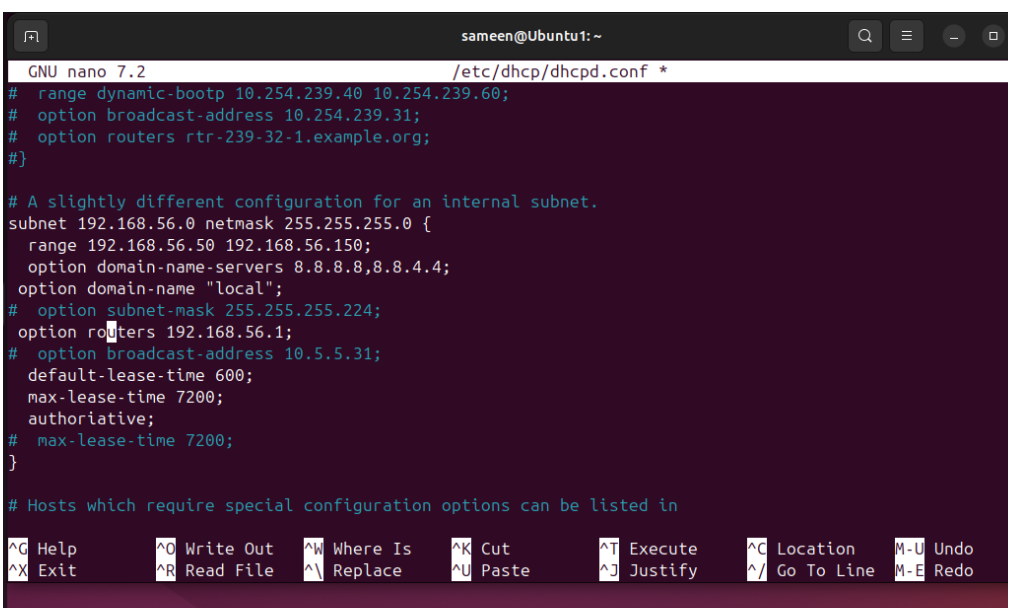
subnet 192.168.56.0 netmask 255.255.255.0 {

range 192.168.56.100 192.168.56.200;

option routers 192.168.56.1;

option domain-name-servers 8.8.8.8, 8.8.4.4;

}

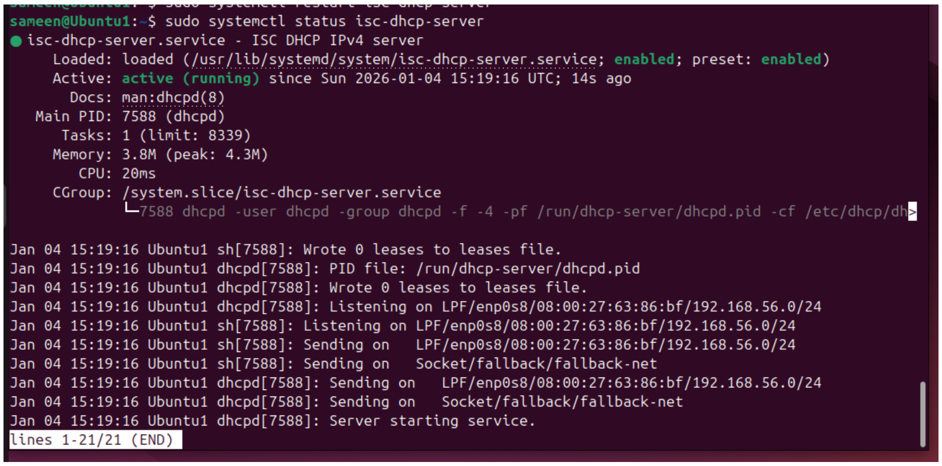


sudo systemctl restart isc-dhcp-server

This command restarts the DHCP server service to apply the new configuration settings.

sudo systemctl status isc-dhcp-server

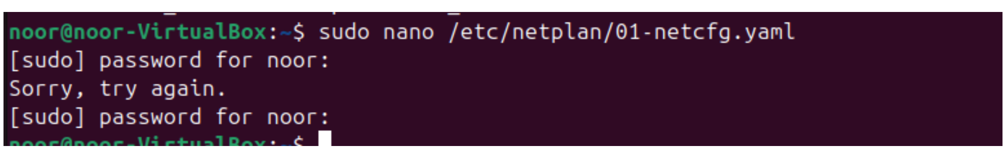
This command verifies whether the DHCP server service is running successfully.



DHCP Client Configuration (Ubuntu 2)

sudo nano /etc/netplan/01-netcfg.yaml

This command opens the Netplan configuration file to enable DHCP on the client's network interfaces.



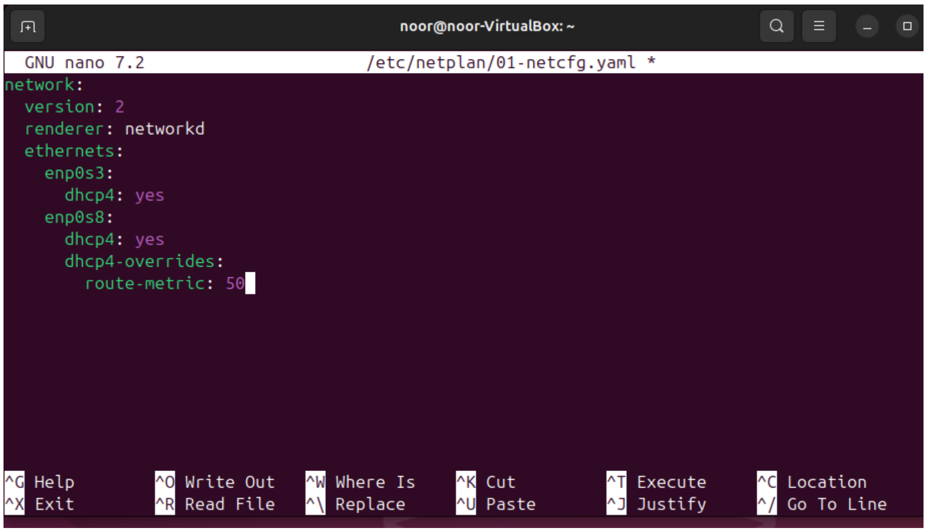
<u>Inside the file paste this:</u>

network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s3:
      dhcp4: yes
    enp0s8:
      dhcp4: yes
      dhcp4-overrides:

route-metric: 50

Purpose:

This configuration enables DHCP on both NAT and Host-Only interfaces and prioritizes the Host-Only network for internal communication.



sudo netplan apply

This command applies the network configuration changes and activates DHCP on the client system.

9. Verification Commands

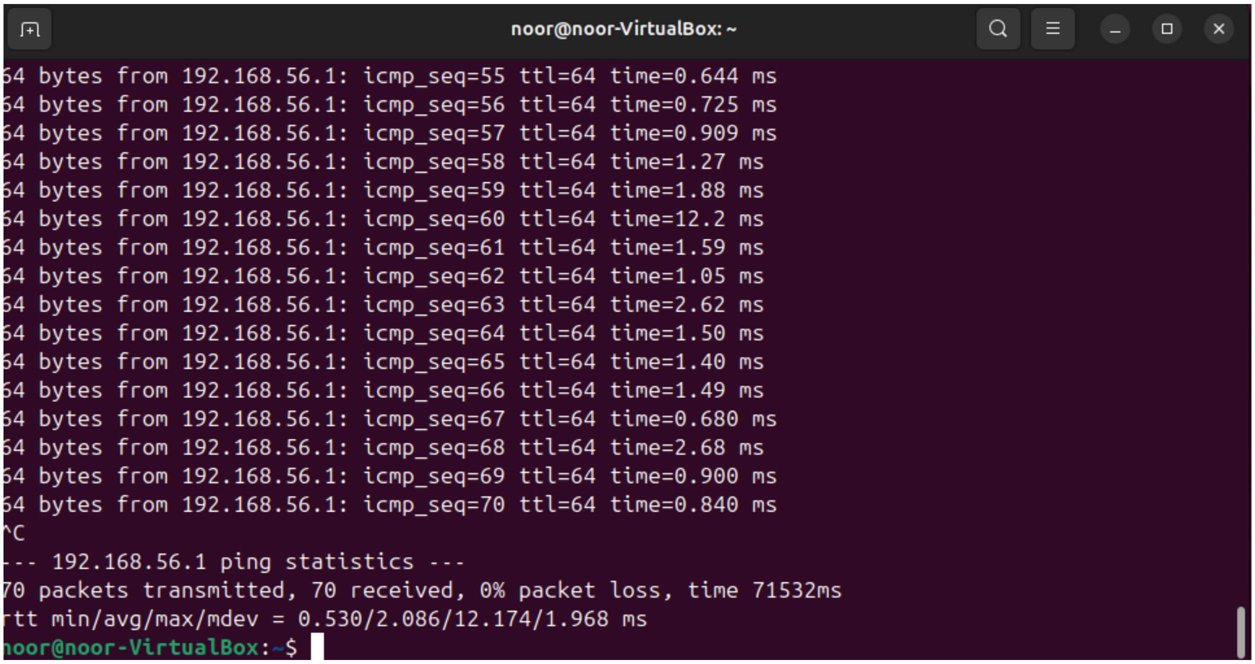
ip a

This command displays the IP addresses assigned to network interfaces, confirming successful DHCP operation.

```
noor@noor-VirtualBox:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qle
n 1000
    link/ether 08:00:27:95:8e:34 brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.6/24 metric 100 brd 192.168.100.255 scope global dynamic enp0s3
       valid_lft 557sec preferred_lft 557sec
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qle
n 1000
    link/ether 08:00:27:5a:81:a2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.4/24 brd 192.168.100.255 scope global enp0s8
       valid_lft forever preferred_lft forever
    inet 192.168.56.104/24 metric 50 brd 192.168.56.255 scope global dynamic enp0s8
       valid_lft 557sec preferred_lft 557sec
```

<u>PING UBUNTU 2(CLIENT) TO UBUNTU 1(SERVER)</u>

```
noor@noor-VirtualBox:~$ ping 192.168.56.1
PING 192.168.56.1 (192.168.56.1) 56(84) bytes of data.
64 bytes from 192.168.56.1: icmp_seq=1 ttl=64 time=2.26 ms
64 bytes from 192.168.56.1: icmp_seq=2 ttl=64 time=1.18 ms
64 bytes from 192.168.56.1: icmp_seq=3 ttl=64 time=0.530 ms
64 bytes from 192.168.56.1: icmp_seq=4 ttl=64 time=1.90 ms
64 bytes from 192.168.56.1: icmp_seq=5 ttl=64 time=1.59 ms
64 bytes from 192.168.56.1: icmp_seq=6 ttl=64 time=6.54 ms
64 bytes from 192.168.56.1: icmp_seq=7 ttl=64 time=11.5 ms
```



Conclusion

A DHCP server was successfully installed and configured on Ubuntu 1 to provide automatic IP addressing. The client system, Ubuntu 2, was able to obtain an IP address dynamically without any manual configuration. Proper network communication between the server and the client was verified through successful ping results. The setup reduced configuration errors and simplified network management. Overall, the objectives of the project were achieved successfully.